

Linking fish physiological biomarkers to habitat integrity in the Atlantic Forest

Biological Conservation OR Neotropical Ichthyology

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The Atlantic Forest is one of the world's most important biodiversity hotspots. Despite occupying only a small fraction of its original distribution, it continues to support extraordinary levels of species richness and endemism. At the same time, it is one of the most threatened tropical ecosystems on Earth due to habitat loss, fragmentation, urban expansion, and agricultural development. The present study evaluated whether biomarkers of physiological stress in fish could detect environmental disturbances occurring within and around the Guaribas Biological Reserve (REBIO Guaribas), a strictly protected area of Atlantic Forest in northeastern Brazil. Despite sites within the administrative boundaries of the REBIO showing no signs alteration, from both environmental variables and biomarkers, significant differences in biomarker responses were observed among sampling sites, indicating that environmental conditions are not homogeneous throughout the stream studied. Elevated levels of all biomarkers at sites located outside or near the boundaries of the protected area suggest exposure to environmental stressors potentially associated with anthropogenic activities. These findings support the use of biochemical and physiological biomarkers as sensitive tools for detecting sublethal effects of environmental degradation, even in landscapes that contain legally protected areas.

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1 Introduction

The Atlantic Forest is one of the world's most important biodiversity hotspots (Myers et al., 2000). Despite occupying only a small fraction of its original distribution, it continues to support extraordinary levels of species richness and endemism (Marques et al., 2021). At the same time, it is one of the most threatened tropical ecosystems on Earth due to habitat loss, fragmentation, urban expansion, and agricultural development (Marques & Grelle, 2021). Conservation efforts in the Atlantic Forest have traditionally focused on terrestrial organisms such as plants, birds, and mammals. However, freshwater ecosystems are equally dependent on forest integrity and are often overlooked in conservation planning (Abilhoa et al., 2011; Gouveia et al., 2017).

Padial et al. (2021) mention a wide range of threats to aquatic systems that can affect the Atlantic Forest, ranging from dam construction and flow alterations to the emerging threat of microplastic contamination (Eerkes-Medrano & Thompson, 2018). Pollution, from pesticides, fertilizers, heavy metals, pharmaceuticals or microplastics (Evans, 1987; Harmon, 2018), come mostly from urban and agricultural sources, and untreated urban sewage is amongst the most common (Costa et al., 2020). In agricultural areas, fish farming, agricultural runoff, and the destruction of riparian forests, remain as important threats to aquatic ecosystem integrity (Padial et al., 2021).

Ecological responses to pollution, often become visible only after environmental impacts have already affected freshwater organisms (Macêdo et al., 2019). Furthermore, the toxicity responsible for behavioral change depends on the organism, time of exposure and particle type, which demonstrates the need for physiological studies, specially in natural environments (Amoatey & Baawain, 2019). For these reasons, physiological biomarkers can provide particularly valuable early-warning indication (Ondei et al., 2020), as physiological, morphological and behavioural changes as species are affected in the ecosystem.

Fish are among the most widely used organisms to evaluate the health of aquatic systems, since their biochemical and physiological changes serve as biomarkers of environmental pollution (Cazenave et al., 2009; Ondei et al., 2020). One of the main physiological functions affected by pollutants is the osmoregulation, and its associated pH control. Both are predominantly done by gills, and are critical to gas exchange, ionic regulation and nitrogen excretion (Evans, 1987). The effects of this action on fish can be measured through a few blood markers, mainly lower levels of Chloride and Sodium ions, obtained by freshwater fish through exchange with internal ions (Evans, 1987), and higher levels of plasmatic proteins (Sabae & Mohamed, 2015). This osmoregulation impairment and consequent decrease in plasma ionic concentration can represent an osmotic challenge for tissues. The hydration level of gills and muscles is another physiological parameter associated with an animal's capacity to cope with environmental pollution and is therefore subject to measurable changes induced by toxic compounds present in the environment. This active control was suggested by Ayrapetyan (2012) as a universal biomarker for detecting pollution in the environment (David et al., 2018; Sabae & Mohamed, 2015).

Another relevant biomarker is the abnormal activity of the multixenobiotic resistance (MXR) phenotype (Bard, 2000; Žaja et al., 2006), an innate gene of many aquatic species that is expressed in organs such as gills, kidneys, blood-brain barrier, liver and pancreas. It is activated by exposure to pollutants and results in the expression of P-glycoproteins (Pgp) in the membrane, which act to remove endogenous or exogenous toxic compounds out of the cell, protecting DNA and inducing toxic compounds excretion (Smital & Kurelec, 1998). Although the MXR activity explains how some species are more resistant to the presence of pollutants, some substances can reverse its effectiveness, being called chemosensitizers (or simply MXR-inhibitors) (Smital et al., 2004). Those compounds can interact with the Pgp activity, altering the mechanism regulation, and saturate or even reverse MXR activity. By saturating the Pgp pump, these chemosensitizers allow other xenobiotics to enter the cell and increase its toxicity, even if the chemosensitizers themselves are not toxic. Precisely because they are mostly organic, natural, primarily harmless and very present in the materials emitted anthropogenic in nature, its detection should also be a priority in environmental risk studies (Kurelec et al., 2000).

In the present study, the hypothesis of markers linked to physiological stress of fish providing evidence of the environmental stress was tested along a headwater stream that originates within a protected area - Biological Reserve, or REBIO, according to BRASIL (2000) - and its course extends beyond its administrative boundaries (MMA, 2003). Given the importance of the Atlantic Forest to the biodiversity (Pereira et al., 2026), we aim with this study, (1) to identify physiological stress biomarkers in fish from a headwater stream, (2) to associate these biomarkers to the ecological health of sites inside and surrounding a protected area of Atlantic Forest. This will enable decision makers to take action on the influence of anthropogenic threats and the ecological consequence of pollution exposure of the buffer zone.

2 Material and Methods

Study area and sampling design. This study was conducted in the Guaribas Biological Reserve (REBIO Guaribas), a protected area of Atlantic Forest in northeastern Brazil. The reserve is located in the municipality of Mamanguape, state of Paraíba, in a heterogeneous landscape influenced by the adjacent Caatinga ecoregion. It was created in 1990 to preserve one of the last remnants of the Atlantic Forest in the region (MMA, 2003). Despite a REBIO being one of the most restricted types of protected areas (BRASIL, 2000), there are a number of human settlements in its buffer zone. The management plan for the REBIO states several rural communities directly surrounding the reserve (Caiana, Pepina, Imbiribeira, Brejinho, Água Fria, João Pereira and Piabuçu), in addition to indigenous human populations (mostly the Potiguaras) (MMA, 2003). These populations are important because of their historic patterns of occupation and land use, but also because their agricultural activities, fishing, modification of the natural vegetation for livestock and overall use of the natural resources from the buffer zone directly affect the conservation of the REBIO. This has been aggravated in recent years by large scale use of water and land for cultivation of monocultures, like

sugarcane (Heinrichs et al., 2017). The surrounding landscape is predominantly agricultural, including coconut plantations, livestock farming, orchards subsistence agriculture. These land uses can act as sources of diffuse pollution, especially through pesticide runoff and domestic effluents, creating a spatial gradient of environmental disturbance (MMA, 2003; Soler, 2004). Thus, the landscape is mainly rural and the land use causes direct impact to aquatic systems, by causing siltation of streams and contaminating groundwater through the use of septic tanks and releasing agricultural chemicals into the soil (Arruda et al., 2013; MMA, 2026)

The climate in the region is classified as As', characterized by a rainy season from February to July and a dry period from October to December (Peel et al., 2007). The average annual temperature varies from 24 to 26 °C, while the annual rainfall varies between 1,750 and 2,000 mm. The hydrographic network is composed of small headwater streams fed by rain, notably the Barro Branco and Caiana streams, both tributaries of the Camaratuba River (Gouveia et al., 2017).

Sampling was conducted along the Barro Branco stream between November 2023 and February 2024, comprising six sampling sites. The sampling sites were distributed longitudinally to capture environmental variability and gradients of human influence. Sites P1 to P3 were located within the reserve, where more preserved conditions were expected, while sites P4 to P6 were situated in the buffer zone adjacent to areas with greater human influence (Figure 1).

For the sake of comparison, we classified all sites into four categories defined *a priori* (Figure 2). Reference site or Type I site, which are the sites that occur within the protected area and are well preserved, with intact riparian forest and no crop fields nearby (at least 1 km distant), free flowing water (0.25 m/s), high dissolved oxygen concentration (6 mg/L) and temperature below 28 °C. Type II sites, are those that occur within the protected area, but are not well preserved, failing in one or more of the water quality criteria and/or having its riparian vegetation removed or managed. Type III sites are those outside the protected area but meet the water quality criteria for a type I site, and type IV sites are those that fall outside the protected area and are also disturbed, not meeting the water quality criteria for a type I site. Sites type III and IV were intentionally chosen not to meet the riparian forest and crop proximity criteria. Since they fall outside of the REBIO they are not expected to have intact riparian forest or to be at least 1 km distant from a crop field. These categories were latter confirmed using environmental data.

The ecophysiological evaluation of human impact on the forest, was mainly guided by the fish species inventory for the REBIO made by Gouveia et al. (2017). This study lists 18 species from five different orders: Characiformes (12 species), Perciformes (3 species), Synbranchiiformes (1 species), Siluriformes (1 species) e Cyprinodontiformes (1 species). From those, two are exotic - *Oreochromis niloticus*, second most farmed fish species in the world and tolerant to osmotic variation, and *Poecilia reticulata*, largely used in mosquito control and resistant to stressful environments (Miranda, 2012). The predominance of small fish inhabiting small basins is thought to lead to low dispersion and consequent isolation of populations (Gouveia et al., 2017), associated with the fact that most species in the study area are phylogenetic close (with the predominance of Characidae) and are widely distributed, offers an appropriate design

to access the entire local ichthyofauna using a few species, namely the genres *Hemigrammus* and *Astyanax*.

Environmental data collection. In order to quantitatively confirm the classification of site category types (I, II, III and IV), environmental characteristics of each site were assessed based on four sets of variables: (a) site morphology, (b) water quality, (c) sediment composition, and (d) marginal habitat structure. Site morphology was characterized by measuring stream width (cm) and depth (cm) from three randomly selected transects. Catchment-scale variables, such as elevation and stream length, were obtained using a handheld GPS and satellite imagery. Water quality was evaluated through physical and chemical variables measured with portable equipment, including temperature (°C) and dissolved oxygen concentration (mg/L). Water turbidity was determined in the laboratory using a turbidimeter, based on water samples collected at each sampling site. Water velocity (m/s) was estimated using the float method by Maitland (1990). Sediment composition and habitat physical structure were assessed following protocols adapted by Medeiros et al. (2008) from Pusey et al. (2004) and Mugodo et al. (2006). These variables were estimated from 9 to 12 one-meter quadrats distributed along the stream margins (terrestrial-aquatic interface). Within each quadrat, the percentage cover of sediment types (mud, sand, cobbles, small gravel, large gravel, rocks, and bedrock) and littoral and subaquatic habitat structures was visually estimated. Habitat structure variables included macrophytes, marginal grasses, leaf litter, attached algae, filamentous algae, overhanging vegetation, submerged vegetation, small woody debris, large woody debris, and root masses.

Three 250 mL replicate samples of water from each site were taken and refrigerated for concentration analysis (g L^{-1}) of ammonia, nitrite, nitrate, orthophosphate (soluble reactive phosphorus) and total phosphorus. Nutrient concentrations were obtained based on Standard Methods for the Examination of Water (1998) and Mackereth et al. (1978).

Fish collection and maintenance. Fish were collected between October and November 2023 along the Barro Branco Stream at six previously established sampling points, three located inside and three outside the Guaribas Biological Reserve, following a methodology adapted from Gouveia et al. (2017). Sampling was carried out during the day using a beach seine net (4 m long, 1.5 m high, and 5 mm mesh) and a hand net (50 cm wide and 5 mm mesh) operated with sweeping movements in shallow waters, according to Medeiros et al. (2010). Sampling effort was standardized across all collection events and sampling points. Fish sampling was performed randomly, therefore the occurrence of the species at each sampling site and the sample size varied according to natural dynamic of fish populations.

Fish were transferred to plastic bags containing water from the collection site and atmospheric oxygen, and transported to the Laboratory of Animal Ecophysiology (LEFA) at the State University of Paraíba. Fish were allowed to acclimated for a few hours to minimise stress from handling and transportation, and the ecophysiological experiments occurred on the same day of catch (Macêdo et al., 2019).

Experimental design. For analyses related to osmoregulatory function, blood samples were obtained from 78 individuals, including 38 fish collected within the reserve and 40 in external areas. The fish were anesthetized with eugenol, and blood aliquots were collected by cardiac puncture using pipettes with heparinized tips. The samples were kept on ice until laboratory processing and subsequently centrifuged for plasma separation. Due to the reduced volume of plasma obtained as a consequence of the small body size of the specimens, chloride ion analysis could not be performed. Branchial and muscle tissues from the same individuals were collected to determine tissue moisture content. Plasma and tissue samples were stored at -20°C until laboratory analysis was performed. The remaining 111 individuals, comprising 54 specimens from locations within the reserve and 57 from external areas, were used for evaluation of the multixenobiotic resistance (MXR) phenotype, following the methodologies described by David et al. (2018); Macêdo et al. (2019); Santos et al. (2017).

Plasmatic protein dosage. For the the determination of plasmatic protein dosage, the plasma samples were defrosted and an aliquot of 2 μL of each was diluted with 8 μL of distilled water (proportion) to perform the total protein dosage, where the dye was also diluted in proportion (BioRad® protein assay) (Bradford, 1976). The intensity of each sample coloration was measured with a microplate reader (SpectraMax i3) at 595 nm wave length, and compared to a standard curve (BSA protein). The results were transferred to a spreadsheet, where the average value in milligrams of protein dosage was calculated for each sampling site.

Moisture content. For the analysis of moisture content, parameter related to the osmoregulatory function, fish tissue samples (gill and muscle) were individually stored in 2 mL eppendorf tubes (previously weighted), initially weighted with the tissue moist, and, after 24 hours inside a 60°C oven, weighted with the dry tissue and the percentual content of tissue water was calculated (David et al., 2018).

MXR phenotype activity. The multixenobiotic resistance activity was analysed through the rhodamine B accumulation assay adapted from Smital & Kurelec (1998). Rhodamine B is a substrate of P-glycoprotein, the molecular basis of the MXR phenotype. Thus, after exposing the fish to this substrate, the analysis of the amount of rhodamine accumulated in the animal's tissues (mainly the gills) reflects the activity of the MXR phenotype: the greater the accumulation, the lower the activity, and the lower the accumulation, the greater the activity. Therefore, specimens of fish from each sampling site (three inside the Conservative Unity and three outside, total of six sampling sites) were transferred to plastic aquariums containing dechlorinated water and rhodamine B at 2,5 μM concentration, staying in this condition for one hour. The aquarium remained under constant aeration and protected from direct light. After exposition, the fish were anesthetized with eugenol and sacrificed for gill sampling. The tissue samples were allocated in Eppendorfs tubes and weighted in an analytical scale. The weight of moist tissue was obtained by subtracting the weight of the previously weighted empty eppendorf tubes. The tissues were then homogenized with 500 μL of distilled water and the supernatant was transferred in triplicate of 100 μL to a 96-well microplate. The fluorescence intensity of the supernatant (corresponding to the intracellular rhodamine B fluorescence in the tissue) was measured in the microplate reader (SectraMax i3) at 544 nm excitation and

590 nm emission. The fluorescence value was then normalized by the moist tissue weight in milligrammes (Macêdo et al., 2019).

Statistical analysis. Environmental variables were assessed for multivariate collinearity, excluding redundant variables and combining those that conveyed similar information to reduce data redundancy, following the approach described by Sokal & Rohlf (1995).

```
1  #MXR23 - ORGANIZANDO DADOS-----
2  #####....-----
3
4  dev.off() #apaga os graficos, se houver algum
5  rm(list=ls(all=TRUE)) #limpa a memória
6  cat("\014") #limpa o console
7  getwd()
8  library(openxlsx)
9
10 ##CARREGANDO MATRIZES BRUTAS-----
11
12 habitat <- read.xlsx("D:/Elvio/OneDrive/MSS/_rebio23-mxr/mxr23_Q/data/rebio23-habitat.xlsx",
13                     rowNames = T,
14                     colNames = T,
15                     sheet = "ambiente",
16                     startRow = 2) #rows = 2:98
17
18 habitat[is.na(habitat)] <- 0
19 habitat
20
21 ### REMOVENDO COLUNAS ZERADAS DE HABITAT-----
22 sum <- colSums(habitat)
23 sum
24 zero_sum <- names(which(colSums(habitat) == 0))
25 zero_sum #nomes das colunas zeradas
26 m_part_cols <- habitat[(colSums(habitat) != 0)] #em != a exclamação inverte o sentido
27 zero_sum2 <- names(which(colSums(m_part_cols) == 0))
28 zero_sum2 #nomes das colunas zeradas
29 sum<-colSums(m_part_cols)
30 sum
31
32 t_grps <- read.xlsx("D:/Elvio/OneDrive/MSS/_rebio23-mxr/mxr23_Q/data/rebio23-habitat.xlsx",
33                    rowNames = T,
34                    colNames = T,
35                    sheet = "grupos",
36                    startRow = 2) #rows = 2:98
```

```

37 t_grps
38
39
40 ##SALVANDO MATRIZES FINAIS----
41
42 write.table(m_part_cols, "m_hab.csv",
43             sep = ";", dec = ".", #"\t",
44             row.names = TRUE,
45             quote = TRUE,
46             append = FALSE)
47 write.table(t_grps, "t_grps.csv",
48             sep = ";", dec = ".", #"\t",
49             row.names = TRUE,
50             quote = TRUE,
51             append = FALSE)
52 t_grps <- read.csv("t_grps.csv",
53                   sep = ";", dec = ".",
54                   row.names = 1,
55                   header = TRUE,
56                   na.strings = NA)
57 m_hab <- read.csv("m_hab.csv",
58                 sep = ";", dec = ".",
59                 row.names = 1,
60                 header = TRUE,
61                 na.strings = NA)
62
63
64 1 #MXR23----
65 2 ####----
66 3
67 4 ##ORGANIZANDO DADOS----
68 5
69 6 dev.off()
70 7 rm(list=ls(all=TRUE))
71 8 cat("\014")
72 9
73 10 t_grps <- read.csv("t_grps.csv",
74                    sep = ";", dec = ".",
75                    row.names = 1,
76                    header = TRUE,
77                    na.strings = NA)
78 11
79 12 m_hab <- read.csv("m_hab.csv",
80                  sep = ";", dec = ".",
81                  row.names = 1,

```

```

18             header = TRUE,
19             na.strings = NA)
20
21 ##CORRELOGRAMA E REMOÇÃO DE VARIÁVEIS REDUNDANTES OU DESNECESSÁRIAS----
22
23 library(psych)
24 colnames(m_hab)
25
26 png("fig-h.hab_pairs.png")
27 pairs.panels(m_hab[,15:24],
28             method = "pearson", # correlation method
29             scale = FALSE, lm = FALSE,
30             hist.col = "#00AFBB", pch = 19,
31             density = TRUE, # show density plots
32             ellipses = TRUE, # show correlation ellipses
33             alpha = 0.5)
34 dev.off()
35
36 cor <- cor(m_hab)
37 cor
38
39 library(corrplot)
40 png("fig-hab_corrplot.png")
41 corrplot(cor, method = "circle")
42 dev.off()
43
44 ##DELETANDO COLINEARES OU INDESEJADAS----
45
46 sink(file = "colineares.txt", append = F, split = T)
47 colnames(m_hab)
48 del_cols <- c("m.Depth_max_cm", "m.Depth_mar_cm")
49 del_cols <- c(del_cols, "n.Ammonia_NH3_ug.L", "n.Nitrite_NO2_ug.L",
50             "n.Nitrate_NO3_ug.L", "n.Orthophosphate_SRP_ug.L",
51             "n.Phosphorus_TP_ug.L")
52 #del_cols <- grep("^n\\.\\.", colnames(m_hab), value = TRUE)
53 m_hab_part <- m_hab[, !(colnames(m_hab) %in% del_cols)]
54
55 ##SOMANDO REDUNDANTES----
56
57 m_hab_part$h.algae <- m_hab_part$h.Algae_f + m_hab_part$h.Algae_a
58 m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
59                             %in% c("h.Algae_f", "h.Algae_a"))]
60 m_hab_part$h.debris <- m_hab_part$h.Debris_l + m_hab_part$h.Debris_s

```

```

61 m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
62                               %in% c("h.Debris_l", "h.Debris_s"))]
63
64 colnames(m_hab_part)
65 m_hab_part
66 sink()
67
68 write.table(m_hab_part, "m_hab_part.csv",
69             sep = ";", dec = ".", #"\\t",
70             row.names = TRUE,
71             quote = TRUE,
72             append = FALSE)
73 m_hab_part <- read.csv("m_hab_part.csv",
74                       sep = ";", dec = ".",
75                       row.names = 1,
76                       header = TRUE,
77                       na.strings = NA)
78
79 1 colnames(m_hab)
80 2 library(PerformanceAnalytics)
81 3 chart.Correlation(m_hab[, c(15:24, 27:31)], histogram=TRUE)
82 4 # ***p<0.001, **p<0.01, *p<0.05, .p<0.10
83
84 5
85 6 library(Hmisc)
86 7 mat <- as.matrix(m_hab[, 15:24])
87 8 res <- rcorr(mat, type = "pearson")
88 9 cor_mat <- res$r
89 10 p_mat <- res$P
90
91 11
92 12 stars <- ifelse(p_mat < 0.001, "***",
93                 ifelse(p_mat < 0.01, "**",
94                 ifelse(p_mat < 0.05, "*", "")))
95 15
96 16 cor_star <- ifelse(p_mat < 0.05,
97                     paste0(round(cor_mat, 2), stars),
98                     NA)
99 19
100 20 cor_star <- matrix(
101 21   cor_star,
102 22   nrow = nrow(cor_mat),
103 23   dimnames = dimnames(cor_mat)
104 24 )
105 25

```

```

26 cor_star[upper.tri(cor_star, diag = TRUE)] <- NA
27 cor_star <- noquote(cor_star)
28 print(cor_star, na.print = "")

```

These variables were subsequently subjected to Principal Component Analysis (PCA) to evaluate multivariate correlations among sites. Prior to analysis, site morphology and water quality variables were square root transformed, whereas sediment composition and the marginal habitat structure (which were measured as percentages) were arcsine square-root transformed after relativization by column total (McCune & Grace, 2002). PCA was performed using the FactoMineR package in R (Lê et al., 2008). All variables were centered and scaled to unit of variance.

```

1  #PCA fviz package----
2  #browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract
3
4  #ORGANIZANDO DADOS----
5
6  dev.off()
7  rm(list=ls(all=TRUE))
8  cat("\014")
9
10 t_grps <- read.csv("t_grps.csv",
11                    sep = ";", dec = ".",
12                    row.names = 1,
13                    header = TRUE,
14                    na.strings = NA)
15 m_hab_part <- read.csv("m_hab_part.csv",
16                       sep = ";", dec = ".",
17                       row.names = 1,
18                       header = TRUE,
19                       na.strings = NA)
20
21 colnames(m_hab_part)
22
23 ###RELATIVIZAÇÕES E TRANSFORMAÇÕES-----
24 library(vegan)
25 library(tidyverse)
26
27 # Selecionando variáveis que começam com "w" ou "m"
28 m_q <- m_hab_part[, grep("^(w|m)", colnames(m_hab_part))]
29
30 # Selecionando variáveis que começam com "s" ou "h"
31 m_h <- m_hab_part[, grep("^(s|h)", colnames(m_hab_part))]

```



```

32
33 # Removendo os dois primeiros caracteres dos nomes das colunas
34 colnames(m_q) <- sub("^..", "", colnames(m_q))
35 colnames(m_h) <- sub("^..", "", colnames(m_h))
36
37 # Transformações
38 m_q <- sqrt(m_q)
39 m_h <- asin(sqrt(decostand(m_h, method = "total", MARGIN = 2)))
40
41 # Juntando tudo em uma única tabela
42 hab_trns <- cbind(m_h, m_q)
43
44 # Salvado m_hab_trns
45 write.table(hab_trns, "m_hab_trns.csv",
46             sep = ";", dec = ".", #"\t",
47             row.names = TRUE,
48             quote = TRUE,
49             append = FALSE)
50 m_hab_trns <- read.table("m_hab_trns.csv",
51                          sep = ";", dec = ".",
52                          row.names = 1,
53                          header = TRUE,
54                          na.strings = NA)

```

All biomarker results were compared with those from Site 1, which served as the reference site for this study. Site 1 was located deep within the protected area, distant from anthropogenic influences, and met the criteria for a Type I site. After ascertaining the non-normality of the study biomarkers, comparisons were performed using one-sided Wilcoxon rank-sum tests to evaluate the hypothesis that biomarker values at each site were significantly greater than the mean observed at the reference site (Zar, 2010). Different fish species were pooled withing sites. All statistical analyses were conducted in the R statistical environment using RStudio (R Core Team, 2017; R Studio Team, 2022).

3 Results

Environmental variables and nutrient concentration. Water velocity ranged from 0 to 0.293 m/s, with the highest values recorded at site 5. Width varied from 1.2 to 14.6 m, while average depth ranged from 13.7 to 44.2 cm. Dissolved oxygen ranged from 5.4 to 8.1 mg/L, and temperature from 25.2 to 27.8 °C. Conductivity varied between 129.0 and 163.3 S/cm, whereas pH ranged from 5.5 to 7.0. Turbidity was generally low, although higher values were observed at sites 5 and 6, reaching 17.5 and 29.6 NTU, respectively. Elevation ranged from 41.5 to 112.8 m a.s.l., while slope varied between 1.0 and 3.0. Mud and sand were

the main substrate components across all sampled sites, with average cover ranging from 30 to 100% and from 0 to 70%, respectively. Habitat structure varied along the stream course. The reference site (site 1) was characterized by high contributions of overhanging vegetation (90%) and exclusively muddy substrates. Site 2 exhibited similar characteristics, whereas site 3 showed reduced habitat complexity. Sites 4–6 displayed greater variation in habitat composition. The habitat structures contributing most overall were overhanging vegetation, debris, and root masses, although these contributions varied among sites. Aquatic macrophytes and littoral grass were particularly abundant at site 4, contributing on average 35.8% and 39.2%, respectively, whereas root masses represented the dominant habitat structure at site 5 (76.7%). Sand cover increased downstream, reaching 70% at site 6, where overhanging vegetation was markedly reduced (Table 1).

```

1  #TABELA DE HABITAT----
2
3  m_hab_part <- read.csv("m_hab_part.csv",
4                        sep = ";", dec = ".",
5                        row.names = 1,
6                        header = TRUE,
7                        na.strings = NA)
8  t_grps <- read.csv("t_grps.csv",
9                    sep = ";", dec = ".",
10                   row.names = 1,
11                   header = TRUE,
12                   na.strings = NA)
13
14  library(dplyr)
15  library(tidyr)
16  m_trab <- m_hab_part %>%
17    rename_with(~ gsub("_", ".", .))
18  m <- m_trab %>%
19    group_by(PontoN = t_grps$PontoN) %>%
20    summarise(across(where(is.numeric),
21                     list(mean = mean, min = min, max = max)),
22              .groups = 'drop') %>%
23    pivot_longer(
24      cols = -c(PontoN),
25      names_to = c("Variable", ".value"),
26      names_sep = "_"
27    )
28
29  m <- as.data.frame(m)
30  m_wide <- m %>%
31    mutate(stat_string = ifelse(Variable == c("m.Vel.m.s"),

```

```

32         paste0(round(mean, 3), " (", round(min, 3), "-", round(max, 3)
33         paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1)
34     unite("Location", PontoN, sep = "_") %>%
35     select(Variable, Location, stat_string) %>%
36     pivot_wider(names_from = Location, values_from = stat_string)
37
38 m_wide
39 m_wide <- as.data.frame(m_wide)
40 m_wide <- m_wide[, c("Variable", "Ponto5", "Ponto6", "Ponto7", "Ponto8", "Ponto9", "Ponto10")]
41 m_wide
42
43 # Salvado m_wide
44 write.table(m_wide, "m_wide_hab.txt",
45           sep = ";", dec = ".", #"\\t",
46           row.names = TRUE,
47           quote = TRUE,
48           append = FALSE)
49 m_wide_hab <- read.table("m_wide_hab.txt",
50           sep = ";", dec = ".",
51           row.names = 1,
52           header = TRUE,
53           na.strings = NA)
54
55 # Exportando dados para Excel
56 library(openxlsx)
57 #write.xlsx(m_wide, file = "tabela de habitat.xlsx", rowNames = FALSE)
58 wb <- loadWorkbook("tabela de habitat.xlsx")
59 writeData(wb, sheet = "Sheet 1", x = m_wide)
60 saveWorkbook(wb, "tabela de habitat.xlsx", overwrite = TRUE)
61
62 # Tabela final ajustada no Excel
63 library(readxl)
64 hab_gtttable <- read_excel(
65   path = "tabela de habitat.xlsx",
66   sheet = "tabela_final",
67   range = "A1:G25") #ajustar para o tamanho da tabela final
68 hab_gtttable
69 library(gt)
70 hab_gtttable <- gt(hab_gtttable)
71 hab_gtttable <- sub_missing(hab_gtttable,
72   columns = everything(), missing_text = "")
73 #hab_gtttable <- fmt_number(hab_gtttable, columns = "F.O.(%)", decimals = 1)
74 hab_gtttable

```

```

75 gtsave(hab_gtttable, "gt-hab_gtttable_xlsx.html")
76 saveRDS(hab_gtttable, "gt-hab_gtttable_xlsx.rds")

1  # Escolher sumário de uma variavel
2  m
3  var <- "w.Temp.C"
4  m[m$Variable == var, "mean"] #cada valor de var
5  summary(m[m$Variable == var, "mean"]) #sumário dos valores de var
6
7  # Escolher sumário de um grupo de variáveis do df m
8  vars <- unique(grep("^h\\.", m$Variable, value = TRUE))
9  summaries <- list() #criam uma lista vazia para guardar os sumários
10 # Loop para cada variável do grupo e guarda em summaries
11 for (var in vars) {
12   summaries[[var]] <- summary(m[m$Variable == var, "mean"])
13 }
14 #var is a temporary variable used in the for loop to iterate through
15 #each variable name that starts with "h."
16
17 summaries
18 summary_table <- do.call(rbind, lapply(summaries, as.data.frame.list))
19 round(summary_table, 2)
20 #sink(file = "summary_h.txt", split = TRUE)
21 round(summary_table[order(summary_table$Mean, decreasing = FALSE), ], 2)
22 #sink()
23
24 # Tabela limpa
25 #summary_table <- cbind(Variable = rownames(summary_table), summary_table)
26 #rownames(summary_table) <- NULL
27 #colnames(summary_table) <- c("Variable", "Min", "Q1", "Median", "Mean", "Q3", "Max")
28 #summary_table

```

Principal Component Analysis (PCA) described the overall environmental structure of the sampling sites and identified the main variables responsible for their spatial differentiation in terms of physicochemical characteristics, channel morphology, substrate composition, and marginal habitat structure Figure 3. The first two principal components explained 57% of the total variance, with the first axis accounting for 31% and the second axis for 26%. The first axis represented a gradient separating sites I and II from site IV. Sites I and II were associated with higher dissolved oxygen concentrations, higher altitude, muddy substrates, and overlying vegetation, while site IV was characterized by greater channel width, average depth, vegetation cover, sandy substrate, and higher turbidity. The second axis reflected the differences in habitat structure and aquatic vegetation. The upper portion of the ordination was associated with algae, macrophytes, and channel width, while the lower portion was related to submerged

vegetation, root mass, water velocity, turbidity, and sandy substrates. Site III occupied an intermediate position along the first axis but was clearly distinguished along the second axis by its association with submerged vegetation, root mass, and higher water velocity. Overall, PCA indicated a marked environmental gradient between the sampling sites, with site IV differing from the upstream sites mainly due to channel morphology and substrate characteristics, while the local habitat structure and aquatic vegetation explained the separation of site III from the other sites.

```
1  #PCA fviz package----
2  #browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract
3
4  #ORGANIZANDO DADOS----
5
6  dev.off()
7  rm(list=ls(all=TRUE))
8  cat("\014")
9
10 t_grps <- read.csv("t_grps.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_hab_trns <- read.csv("m_hab_trns.csv",
16                      sep = ";", dec = ".",
17                      row.names = 1,
18                      header = TRUE,
19                      na.strings = NA)
20
21 #PCA----
22
23 library("FactoMineR")
24 library("factoextra")
25
26 pca <- PCA(m_hab_trns, scale.unit = TRUE, ncp = 5, graph = TRUE)
27 print(pca)
28
29 eig.val <- get_eigenvalue(pca)
30 eig.val
31
32 sink(file = "cor_matrix.txt", append = FALSE)
33 pca$var$cor
34 sink()
35
```

```

36 fviz_eig(pca, addlabels = TRUE, ylim = c(0,30))
37
38 var <- get_pca_var(pca)
39 var
40
41 fviz_pca_var(pca, col.var = "black")
42
43 library("corrplot")
44 corrplot(var$cos2, is.corr=FALSE)
45
46 fviz_cos2(pca, choice = "var", axes = 1:2)
47
48 fviz_pca_var(pca, col.var = "cos2",
49             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
50             repel = TRUE # Avoid text overlapping
51 )
52
53 fviz_pca_var(pca, alpha.var = "cos2")
54
55 corrplot(var$contrib, is.corr=FALSE)
56
57 fviz_contrib(pca, choice = "var", axes = 1, top = 10)
58 fviz_contrib(pca, choice = "var", axes = 2, top = 10)
59 fviz_contrib(pca, choice = "var", axes = 1:2, top = 10)
60
61 fviz_pca_var(pca, col.var = "contrib",
62             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07")
63 )
64 fviz_pca_var(pca, alpha.var = "contrib")
65
66 ##GROUPING BY KMEANS----
67
68 set.seed(123)
69 res.km <- kmeans(var$coord, centers = 3, nstart = 25)
70 grp <- as.factor(res.km$cluster)
71 # Color variables by groups
72 fviz_pca_var(pca, col.var = grp,
73             palette = c("#0073C2FF", "#EFC000FF", "#868686FF"),
74             legend.title = "Cluster")
75
76 ind <- get_pca_ind(pca)
77 ind
78 ind$contrib

```

```

79
80 ##BIPLOTS----
81
82 fviz_pca_ind(pca)
83
84 fviz_pca_ind(pca, col.ind = "cos2",
85               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
86               repel = TRUE #avoid text overlapping (slow if many points)
87 )
88
89 fviz_pca_ind(pca, pointsize = "cos2",
90               pointshape = 21, fill = "#E7B800",
91               repel = TRUE #avoid text overlapping (slow if many points)
92 )
93
94 fviz_pca_ind(pca, col.ind = "cos2", pointsize = "cos2",
95               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
96               repel = TRUE #avoid text overlapping (slow if many points)
97 )
98
99 fviz_cos2(pca, choice = "ind")
100
101 fviz_contrib(pca, choice = "ind", axes = 1:2)
102
103 # Create a random continuous variable of length 23,
104 # Same length as the number of active individuals in the PCA
105 t_grps <- t_grps[rownames(pca$ind$coord), ]
106 set.seed(123)
107 my.cont.var <- t_grps$Ref_site
108 my.cont.var
109 # Color individuals by the continuous variable
110 fviz_pca_ind(
111   pca,
112   habillage = my.cont.var,
113   palette = c("blue", "yellow", "red", "pink"),
114   repel = FALSE
115 )
116
117 fviz_pca_ind(pca,
118               geom.ind = "point", #show points only (nbut not "text")
119               col.ind = as.factor(my.cont.var), #color by groups
120               palette = "grey",
121               addEllipses = TRUE, #concentration ellipses

```

```

122         legend.title = "Groups"
123     )
124
125     length(unique(my.cont.var))
126     fviz_pca_ind(pca,
127                 geom.ind = "point",
128                 col.ind = t_grps$Ref_site,
129                 palette = "grey",
130                 addEllipses = TRUE, ellipse.type = "confidence",
131                 mean.point = TRUE,
132                 legend.title = "Groups")
133
134     fviz_pca_biplot(pca, repel = TRUE,
135                    col.var = "#2E9FDF", #variables color
136                    col.ind = "#696969"  #individuals color
137    )
138
139     fviz_pca_biplot(pca,
140                    col.ind = t_grps$Ref_site, palette = "jco",
141                    addEllipses = TRUE, ellipse.type = "euclid",
142                    label = "var",
143                    col.var = "black", repel = TRUE,
144                    legend.title = "Species")
145
146     ### FILTRANDO AS VARIÁVEIS
147     var_coord <- pca$var$coord[,1:2]
148     keep_vars <- apply(abs(var_coord) > 0.5,1,any)
149     vars_to_keep <- rownames(var_coord)[keep_vars]
150     pca_filt <- pca
151     pca_filt$var$coord <- pca$var$coord[vars_to_keep,,drop=FALSE]
152     pca_filt$var$cos2 <- pca$var$cos2[vars_to_keep,,drop=FALSE]
153     pca_filt$var$contrib <- pca$var$contrib[vars_to_keep,,drop=FALSE]
154
155     ### AJUSTE MANUAL DAS COORDENADAS DOS VETORES
156     var_coords <- as.data.frame(pca_filt$var$coord)
157     var_coords$Dim.1 <- var_coords$Dim.1*4.2
158     var_coords$Dim.2 <- var_coords$Dim.2*4.2
159
160     ## Caso queira editar manualmente
161     #fix(var_coords)
162     #write.table(var_coords,"coords_var.txt")
163
164     var_coords <- read.table("coords_var.txt",

```



```

165         header=TRUE,
166         row.names=1)
167
168     ### AJUSTE MANUAL DOS INDIVÍDUOS
169     ind_coords <- as.data.frame(pca_filt$ind$coord)
170
171     ## Caso queira editar manualmente
172     #fix(ind_coords)
173     #write.table(ind_coords, "coords_ind.txt")
174
175     ind_coords <- read.table("coords_ind.txt",
176                             header=TRUE,
177                             row.names=1)
178
179     #####
180     ### GRÁFICO FINAL
181     #####
182
183     library(factoextra)
184     library(ggplot2)
185
186     t_grps$Ref_site <- as.factor(t_grps$Ref_site)
187
188     #png("fig-hab_PCA_fviz.png", width=11, height=9, units="in", res=400)
189
190     library(ggplot2)
191
192     ggplot() +
193
194         geom_hline(yintercept = 0, linetype = 2) +
195         geom_vline(xintercept = 0, linetype = 2) +
196
197         geom_point(
198             data = ind_coords,
199             aes(Dim.1, Dim.2, shape = t_grps$Ref_site),
200             size = 3,
201             colour = "black"
202         ) +
203
204         geom_segment(
205             data = var_coords,
206             aes(x = 0, y = 0,
207                 xend = Dim.1,

```

```

208     yend = Dim.2),
209     colour = "red",
210     arrow = arrow(length = unit(0.25,"cm"))
211 ) +
212
213 geom_text(
214     data = transform(
215         var_coords,
216         nudge_y = ifelse(Dim.2 >= 0, 0.25, -0.25)
217     ),
218     aes(
219         x = Dim.1,
220         y = Dim.2 + nudge_y,
221         label = rownames(var_coords)
222     ),
223     colour = "red",
224     size = 4,
225     nudge_x = 0.19
226 ) +
227
228 scale_shape_manual(values = c(15,16,0,1,2,17)) +
229
230 labs(
231     x = paste0("Dim1 (", round(pca$eig[1,2],1), "%)"),
232     y = paste0("Dim2 (", round(pca$eig[2,2],1), "%)"),
233     shape = "Sites"
234 ) +
235
236 coord_equal() +
237 theme_minimal()
238
239 #dev.off()

```

Averaged nutrient concentrations varied among sampling sites (Table 2). Ammonia concentrations ranged from 54.8 $\mu\text{g L}^{-1}$ at the reference site (Site 1) to 243.7 $\mu\text{g L}^{-1}$ at Site 6, the highest value recorded during the study. Elevated ammonia concentrations were also observed at Sites 4 (81.6 $\mu\text{g L}^{-1}$) and 5 (80.8 $\mu\text{g L}^{-1}$), whereas Sites 1–3 exhibited comparatively lower concentrations (54.8–72.0 $\mu\text{g L}^{-1}$). Nitrite concentrations were zero at all sampling sites. Nitrate concentrations ranged from 5.5 to 18.8 $\mu\text{g L}^{-1}$, with the highest values recorded at Sites 4 and 5. Orthophosphate was detected only at Site 3 (1.0 $\mu\text{g L}^{-1}$), whereas total phosphorus ranged from undetectable levels at Site 1 to 50.1 $\mu\text{g L}^{-1}$ at Site 3, with relatively high concentrations also recorded at Site 6 (38.7 $\mu\text{g L}^{-1}$).

```

1  #TABELA DE NUTRI----
2  #ORGANIZANDO DADOS----
3
4  dev.off()
5  rm(list=ls(all=TRUE))
6  cat("\014")
7
8  library(openxlsx)
9  nutri <- read.xlsx("D:/mxr23_Q/data/rebio23-habitat.xlsx",
10                    rowNames = T,
11                    colNames = T,
12                    sheet = "nutri",
13                    rows = 2:20) #rows = 2:98
14
15  t_grps <- read.csv("t_grps.csv",
16                    sep = ";", dec = ".",
17                    row.names = 1,
18                    header = TRUE,
19                    na.strings = NA)
20
21  library(dplyr)
22  library(tidyr)
23  t_grps <- t_grps[seq_len(nrow(nutri)), , drop = FALSE]
24  m_trab <- nutri %>%
25    rename_with(~ gsub("_", ".", .))
26  m <- m_trab %>%
27    group_by(PontoN = t_grps$PontoN) %>%
28    summarise(across(where(is.numeric),
29                      list(mean = mean, min = min, max = max)),
30              .groups = 'drop') %>%
31    pivot_longer(
32      cols = -c(PontoN),
33      names_to = c("Variable", ".value"),
34      names_sep = "_"
35    )
36
37  m <- as.data.frame(m)
38  colnames(m_trab)
39  m_wide <- m %>%
40    mutate(stat_string = ifelse(Variable == c("Ammonia.NH3.ug.L"),
41                                   paste0(round(mean, 3), " (", round(min, 3), "-", round(max, 3)
42                                   paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1)
43    unite("Location", PontoN, sep = "_") %>%

```

```

44   select(Variable, Location, stat_string) %>%
45   pivot_wider(names_from = Location, values_from = stat_string)
46
47   m_wide
48   m_wide <- as.data.frame(m_wide)
49   m_wide <- m_wide[, c("Variable", "Ponto5", "Ponto6", "Ponto7", "Ponto8", "Ponto9", "Ponto10")]
50   m_wide
51
52   # Salvado m_wide
53   write.table(m_wide, "m_wide_nutri.txt",
54               sep = ";", dec = ".", #"\\t",
55               row.names = TRUE,
56               quote = TRUE,
57               append = FALSE)
58   m_wide_nutri <- read.table("m_wide_nutri.txt",
59                             sep = ";", dec = ".",
60                             row.names = 1,
61                             header = TRUE,
62                             na.strings = NA)
63
64   # Exportando dados para Excel
65   library(openxlsx)
66   #write.xlsx(m_wide, file = "tabela de nutrientes.xlsx", rowNames = FALSE)
67   wb <- loadWorkbook("tabela de nutrientes.xlsx")
68   writeData(wb, sheet = "Sheet 1", x = m_wide)
69   saveWorkbook(wb, "tabela de nutrientes.xlsx", overwrite = TRUE)
70
71   # Tabela final ajustada no Excel
72   library(readxl)
73   nutri_gtable <- read_excel(
74     path = "tabela de nutrientes.xlsx",
75     sheet = "tabela_final",
76     range = "A1:G6") #ajustar para o tamanho da tabela final
77   nutri_gtable
78   library(gt)
79   nutri_gtable <- gt(nutri_gtable)
80   nutri_gtable <- sub_missing(nutri_gtable,
81                               columns = everything(), missing_text = "")
82   nutri_gtable
83   gtsave(nutri_gtable, "gt-nutri_gtable.xlsx.html")
84   saveRDS(nutri_gtable, "gt-nutri_gtable.xlsx.rds")

```

Fish community. A total of six fish species were collected under a standardized sampling

effort, so species representation reflected their natural abundances in the study area. Among the families recorded, Characidae was the richest, comprising five species (*Astyanax bimaculatus*, *Astyanax fasciatus*, *Hemigrammus marginatus*, *Hemigrammus rodwayi*, and *Hemigrammus unilineatus*), whereas Crenuchidae was represented by a single species, *Characidium bimaculatum*, from which only one specimen was collected. *Hemigrammus unilineatus* was the most abundant species, representing 66.3% of all individuals analyzed, followed by *Astyanax fasciatus* (14.8%) and *Hemigrammus rodwayi* (11.2%). *Astyanax bimaculatus* accounted for 4.6% of the specimens, whereas *Hemigrammus marginatus* and *Characidium bimaculatum* represented only 2.5% and 0.5%, respectively. Since species representation in the physiological analyses reflected their natural abundances, the number of individuals available was divided for each subsequent biomarker analysis, in order to reflect the natural composition of the fish assemblage. Of the total fish collected, 118 specimens were submitted to the rhodamine B accumulation assay and 78 were used for total plasma protein and moisture content analyses (Figure 4). Given its low count, *Characidium bimaculatum* was removed from all biomarker analyses.

```

1 dev.off() #apaga os graficos, se houver algum
2 rm(list=ls(all=TRUE)) #limpa a memória
3 cat("\014") #limpa o console
4
5 getwd()
6 library(openxlsx)
7 mxr23 <- read.xlsx("D:/Elvio/OneDrive/MSS/_rebio23-mxr/mxr23_Q/data/rebio23-peixes_5-10.xlsx",
8                   rowNames = T,
9                   colNames = T,
10                  sheet = "peixes-fisio")
11 mxr23#[1:5,1:5] mostra apenas as linhas e colunas de 1 a 5.
12
13 ##### ARRUMANDO OS DADOS #####
14
15 mxr23_part <- mxr23[!row.names(mxr23) %in% c("4-9-85", #NA
16                                             "1-10-7"),] #As.bim grande
17
18 mxr23_part[c("1-10-4", "1-10-5"), c("Rodamina_rfu_mg", "TH_musculo_p",
19                                     "TH_branquia_p", "Proteinas_Totais_mg"
20                                     )] <- NA
21 mxr23_part <- mxr23_part[mxr23_part$Especie != "Characidium bimaculatum", ]
22
23 write.table(mxr23_part, "mxr23_part.csv",
24             sep = ";", dec = ".", #"t",
25             row.names = TRUE,
26             quote = TRUE,
27             append = FALSE)

```

```

28 data <- read.csv("mxr23_part.csv",
29                 sep = ";", dec = ".",
30                 row.names = 1,
31                 header = TRUE,
32                 na.strings = NA)
33
34 ##### RESUMO DOS DADOS #####
35
36 library(dplyr)
37 library(tidyr)
38
39 grouped_data <- group_by(data, Espécie, Amostra_tipo)
40 summarised_data <- summarise(grouped_data, count = n())
41 species_count_table <- spread(summarised_data, Amostra_tipo, count, fill = 0)
42 print(species_count_table)
43
44 species_count_table <- species_count_table %>%
45   rowwise() %>%
46   mutate(N = sum(c_across(where(is.numeric))))
47
48 resumo <- as.data.frame(species_count_table)
49 resumo
50
51 rownames(resumo) <- resumo[,1] #tem que ser um df
52 resumo[,1] <- NULL
53
54 soma <- apply(resumo, 2, sum)
55 soma
56
57 soma_row <- as.data.frame(t(soma))
58 rownames(soma_row) <- "Soma"
59
60 resumo <- rbind(resumo, soma_row)
61 resumo
62
63 ##### RESUMO SPP POR UA #####
64
65 resumo2 <- data %>%
66   group_by(Site = Site, Espécie = Espécie, Amostra_tipo = Amostra_tipo) %>%
67   summarise(Count = n()) %>%
68   arrange(Site, Espécie, Amostra_tipo)
69
70 resumo2 <- as.data.frame(resumo2)

```

```

71  resumo2
72
73  write.table(resumo2, file = "resumo.csv", row.names = T, sep = "\t")
74  read.table("resumo.csv", check.names = F)
75
76  resumo2_wide <- resumo2 %>%
77    pivot_wider(names_from = c(Site, Amostra_tipo), values_from = Count, values_fill = list(Count = 0))
78  resumo2_wide <- as.data.frame(resumo2_wide)
79  resumo2_wide
80
81  # Create the gt table
82  library(gt)
83  gt_table <- resumo2_wide %>%
84    gt() %>%
85    tab_header(
86      title = "Species Count by Site and Sample Type"
87    ) %>%
88    cols_label(
89      Especie = "Species"
90    ) %>%
91    fmt_number(
92      columns = everything(),
93      decimals = 0
94    ) %>%
95    cols_width(
96      everything() ~ px(100)
97    ) %>%
98    opt_table_outline()
99
100 gt_table
101
102 ##### GRÁFICO POR N #####
103
104 # Filtrar por Amostra_tipo "xx"
105 filtered_Amostra_tipo <- data %>%
106   filter(Amostra_tipo != "xx")
107 data <- filtered_Amostra_tipo
108
109 # Count occurrences of each species by Amostra_tipo and Site
110 species_count <- data %>%
111   group_by(Site, Especie, Amostra_tipo) %>%
112   summarise(Count = n(), .groups = 'drop')
113

```

```

114 # Plot the counts
115 library(ggplot2)
116
117 counts <- ggplot(species_count, aes(x = Amostra_tipo, y = Count, fill = Especie)) +
118   geom_bar(stat = "identity", position = position_dodge()) +
119   facet_wrap(
120     ~ Site,
121     labeller = labeller(
122       Site = c(
123         "B-P05" = "Site 1",
124         "B-P06" = "Site 2",
125         "B-P07" = "Site 3",
126         "B-P08" = "Site 4",
127         "B-P09" = "Site 5",
128         "B-P10" = "Site 6"
129       )
130     )
131   ) +
132   scale_x_discrete(
133     labels = c(
134       "rd" = "MXR",
135       "th" = "Moisture"
136     )
137   ) +
138   scale_fill_grey(start = 0.2, end = 0.8) +
139   labs(title = "",
140        x = "Sample use",
141        y = "Count",
142        fill = "Species") +
143   theme_minimal() +
144   theme(axis.text.x = element_text(angle = 45, hjust = 1))
145 counts
146 ggsave(counts, dpi = 300, filename = "fig-counts.png", bg = "white")
147
148
149 # Count occurrences of each species by Amostra_tipo and Site
150 species_count <- data %>%
151   group_by(Site, Especie, Amostra_tipo) %>%
152   summarise(Count = n(), .groups = 'drop')
153 # Plot the counts with species on the x-axis
154 ggplot(species_count, aes(x = Especie, y = Count, fill = Amostra_tipo)) +
155   geom_bar(stat = "identity", position = position_dodge()) +
156   facet_wrap(~ Site) +

```



```

157 labs(title = "Count of Each Species per Amostra_tipo for Each Site",
158       x = "Species",
159       y = "Count") +
160 theme_minimal() +
161 theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

Plasmatic protein dosage. The number of plasmatic protein dosages (n=44) was smaller than the total number of fish collected for this part of the analysis (n=78), because of the difficulty on collecting blood from smaller specimens and the need to collect a minimum amount of plasma (Figure 5).

```

1 dev.off()
2 rm(list=ls(all=TRUE))
3 cat("\014")
4
5 data <- read.csv("mxr23_part.csv",
6                 sep = ";", dec = ".",
7                 row.names = 1,
8                 header = TRUE,
9                 na.strings = NA)
10
11 table(data$Especie, is.na(data$Proteinas_Totais_mg))
12
13 ##### GRAFICO POR ANÁLISE #####
14
15 #N #data <- data %>% mutate(N = 1)
16 #RPC
17 #Rodamina_rfu_mg
18 #TH_musculo_p
19 #TH_branquias_p
20 #Proteinas_Totais_mg
21
22 # Convert the column to numeric, replace commas with dots for proper conversion
23 data$Proteinas_Totais_mg <- as.numeric(gsub(",", ".", data$Proteinas_Totais_mg))
24 VAR <- deparse(substitute(data$Proteinas_Totais_mg))
25 var <- "Proteinas_Totais_mg" %>% rlang::sym()
26 str(data)
27
28 # Calculate the average for each combination of species, Amostra_tipo, and Site
29 average <- summarise(
30   group_by(data, Site, Especie, Amostra_tipo),
31   Average = mean(!var, na.rm = TRUE),
32   n = sum(!is.na(!var)),

```

```

33 SE = ifelse(
34   sum(!is.na(!var)) > 1,
35   sd(!var, na.rm = TRUE) / sqrt(sum(!is.na(!var))),
36   0
37 ),
38 .groups = "drop"
39 )
40 average <- average[
41   average$Amostra_tipo == "th" & !is.nan(average$Average),
42 ]
43 # Plot the average
44 ggplot(average, aes(x = Espécie, y = Average, fill = Amostra_tipo)) +
45   geom_bar(stat = "identity", position = position_dodge()) +
46   facet_wrap(~ Site) +
47   labs(title = paste("Average POR ANÁLISE para", VAR, "per Species and Amostra_tipo for Each
48     x = "Species",
49     y = paste("Average", VAR)) +
50   theme_minimal() +
51   theme(axis.text.x = element_text(angle = 45, hjust = 1))
52
53 ##### PLOT FINAL #####
54
55 prot_tots <-
56   ggplot(average, aes(x = Espécie, y = Average, fill = Amostra_tipo)) +
57   geom_bar(stat = "identity", position = position_dodge(), fill = "grey70") +
58   geom_errorbar(
59     aes(
60       ymin = Average - SE,
61       ymax = Average + SE
62     ),
63     width = 0.2
64   ) +
65   geom_text(
66     aes(
67       y = 10,
68       label = paste0("n=", n)
69     ),
70     size = 3
71   ) +
72   facet_wrap(
73     ~ Site,
74     labeller = labeller(
75       Site = c(

```

```

76     "B-P05" = "Site 1",
77     "B-P06" = "Site 2",
78     "B-P07" = "Site 3",
79     "B-P08" = "Site 4",
80     "B-P09" = "Site 5",
81     "B-P10" = "Site 6"
82   )
83 )
84 ) +
85   scale_fill_grey(start = 0.2, end = 0.8) +
86   labs(title = "",
87        x = "Species",
88        y = "Plasmatic protein dosage (mg)",
89        fill = "Species") +
90   theme_grey() +
91   theme(axis.text.x = element_text(angle = 45, hjust = 1),
92         legend.position = "none")
93 prot_tots
94 ggsave(prot_tots, dpi = 300, filename = "fig-prot_tots.png", bg = "white")

1  ##### GRÁFICO POR ESPÉCIE #####
2
3  # Filter out the species Astyanax bimaculatus and Astyanax fasciatus
4
5  filtered_data <- data
6
7  filtered_data <- data %>%
8    filter(!Especie %in% c("Astyanax bimaculatus",
9                          "Astyanax fasciatus",
10                         "Characidium bimaculatum")) #EXCLUI USANDO O "!"
11 filtered_data <- data %>%
12   filter(Especie %in% c("Hemigrammus unilineatus")) #ESCOLHE
13
14 # Calculate the average for each combination of Amostra_tipo, Especie, and Site
15 average <- filtered_data %>%
16   group_by(Site, Amostra_tipo, Especie) %>%
17   summarise(Average = mean(!var, na.rm = TRUE), .groups = 'drop')
18 # Plot the average RPC with Amostra_tipo on the x-axis
19 ggplot(average, aes(x = Amostra_tipo, y = Average, fill = Especie)) +
20   geom_bar(stat = "identity", position = position_dodge()) +
21   facet_wrap(~ Site) +
22   labs(title = paste("Average", VAR, "per Amostra_tipo for Each Site"),
23        x = "Amostra Tipo",

```

```

24     y = paste("Average", VAR)) +
25     theme_minimal() +
26     theme(axis.text.x = element_text(angle = 45, hjust = 1))
27
28 ##### CORRELAÇÕES #####
29
30 data <- filtered_data
31
32 colnames(data)
33 cor <- data %>%
34   select(Ponto, Ref_site, Compr.total_mm, Compr.padrao_mm, Altura_mm,
35     Peso_g, RPC, Rodamina_rfu_mg, TH_branquia_p, TH_musculo_p,
36     Proteinas_Totais_mg)
37 str(cor)
38 cor[] <- lapply(cor, function(x) as.numeric(gsub(",", ".", x)))
39
40 plot(cor)
41 plot(cor[,3:7])
42
43 cor(cor,method="pearson")
44 cor(cor[,1:3], method="spearman")
45
46 library(psych)
47 pairs.panels(cor[,8:11],
48   method = "pearson", # correlation method
49   scale = FALSE, lm = FALSE,
50   hist.col = "#00AFBB", pch = 19,
51   density = TRUE, # show density plots
52   ellipses = TRUE, # show correlation ellipses
53   alpha = 0.5
54 )
55
56 cor %>%
57   with(cor.test(RPC, Peso_g, method = "pearson"))
58
59 with(cor, plot(scale(RPC), scale(Peso_g)))
60 with(cor, plot(RPC, Peso_g))
61
62 plot((m_part$"Overh_veg" - mean(m_part$"Overh_veg")) / sd(m_part$"Overh_veg"))
63 #plot((m_trns$"m.elev" - mean(m_trns$"m.elev")) / sd(m_trns$"m.elev"))
64 with(cor, plot((RPC - mean(Peso_g)) / sd(Peso_g)))
65 with(cor, plot((RPC - mean(RPC)) / sd(RPC),
66   (Peso_g - mean(Peso_g)) / sd(Peso_g)))

```

Plasma total protein concentrations differed between sampling sites when compared with the reference site (Site 1). Mean protein concentrations were significantly higher at the sites 4, 5 and 6 (91.25 ± 5.68 , 74.22 ± 3.31 and 74.79 ± 6.55 mg, respectively) than the reference site (33.53 ± 0.32 mg; Wilcoxon rank-sum test, $p < 0.02$). In contrast, no significant increase was observed at Sites 2 and 3 (29.41 ± 1.59 and 33.75 ± 0.80 mg, respectively; $p > 0.4$). The highest mean protein concentration was recorded at Site 4, representing an approximately 2.7-fold increase relative to the reference site.

```

1 dev.off()
2 rm(list=ls(all=TRUE))
3 cat("\014")
4
5 data <- read.csv("mxr23_part.csv",
6                 sep = ";", dec = ".",
7                 row.names = 1,
8                 header = TRUE,
9                 na.strings = NA)
10
11 table(data$Especie, is.na(data$Proteinas_Totais_mg))
12 var <- "Proteinas_Totais_mg"
13
14 # Reference site
15 ref <- data[[var]][
16   data$Site == "B-P05" &
17   !is.na(data[[var])]
18 ]
19
20 # Sites to compare
21 sites <- setdiff(unique(data$Site), "B-P05")
22
23 # Run tests
24 results <- data.frame()
25
26 for(site in sites){
27
28   test_data <- data[[var]][
29     data$Site == site &
30     !is.na(data[[var])]
31   ]
32
33   tt <- t.test(
34     test_data,
35     ref,

```

```

36     alternative = "greater"
37   )
38
39   results <- rbind(
40     results,
41     data.frame(
42       Site = site,
43       Mean_Site = mean(test_data),
44       SD_Site = sd(test_data),
45       SE_Site = sd(test_data) / sqrt(length(test_data)),
46       Mean_Ref = mean(ref),
47       SD_Ref = sd(ref),
48       SE_Ref = sd(ref) / sqrt(length(ref)),
49       N_Site = length(test_data),
50       N_Ref = length(ref),
51       t = unname(tt$statistic),
52       df = unname(tt$parameter),
53       p = tt$p.value
54     )
55   )
56 }
57
58 results
59
60 results$Significant <- ifelse(results$p < 0.05, "Yes", "No")
61
62 results$p <- ifelse(
63   results$p < 0.001,
64   "<0.001",
65   sprintf("%.3f", results$p)
66 )
67 #sink("t-tests_prot_tots.txt", append = TRUE)
68 results
69 #sink()
70
71 ##### NÃO-PARAMETRICO #####
72
73 results_w <- data.frame()
74
75 for(site in sites){
76
77   test_data <- data[[var]][
78     data$Site == site &

```

```

79     !is.na(data[[var]])
80   ]
81
82   wt <- wilcox.test(
83     test_data,
84     ref,
85     alternative = "greater"
86   )
87
88   results_w <- rbind(
89     results_w,
90     data.frame(
91       Site = site,
92       W = unname(wt$statistic),
93       p = wt$p.value
94     )
95   )
96 }
97
98 results_w$Significant <- ifelse(results_w$p < 0.05, "Yes", "No")
99
100 results_w$p <- ifelse(
101   results_w$p < 0.001,
102   "<0.001",
103   sprintf("%.3f", results_w$p)
104 )
105
106 #sink("t-tests_np.txt", append = FALSE)
107 results_w
108 #sink()

```

Moisture content. Gill and muscle tissues moisture content showed similar results, with gill moisture concentrations being significantly higher than those observed at the reference site (Site 1; $82.82 \pm 1.75\%$) at Sites 5 and 6 (mean values of $90.13 \pm 1.19\%$ and $93.89 \pm 0.98\%$, Wilcoxon rank-sum test; $p < 0.004$). No significant increases were detected at Sites 2, 3, or 4 ($84.35 \pm 1.60\%$, $85.63 \pm 1.94\%$, and $82.84 \pm 0.75\%$, respectively; $p > 0.1$ in all cases).

The muscle moisture concentrations at the reference site averaged $80.31 \pm 1.48\%$. Significantly higher values were observed at Sites 4, 5 and 6, where mean moisture concentrations reached $82.38 \pm 0.62\%$ and $84.29 \pm 0.88\%$, $86.02 \pm 2.49\%$, respectively (Wilcoxon rank-sum test; $p = 0.04$). In contrast, moisture concentrations at Sites 2 and 3 ($80.11 \pm 0.66\%$ and $80.58 \pm 0.94\%$) did not differ significantly from those at the reference site ($p > 0.39$) (Figure 6).

```

1 dev.off()

```

```

2  rm(list=ls(all=TRUE))
3  cat("\014")
4
5  data <- read.csv("mxr23_part.csv",
6                  sep = ";", dec = ".",
7                  row.names = 1,
8                  header = TRUE,
9                  na.strings = NA)
10
11  table(data$Especie, is.na(data$TH_branquia_p))
12  table(data$Especie, is.na(data$TH_musculo_p))
13
14  ##### GRAFICO POR ANÁLISE #####
15
16  #N #data <- data %>% mutate(N = 1)
17  #RPC
18  #Rodamina_rfu_mg
19  #TH_musculo_p
20  #TH_branquias_p
21  #Proteinas_Totais_mg
22
23  library(dplyr)
24  library(tidyr)
25  library(ggplot2)
26
27  # Convert the column to numeric, replace commas with dots for proper conversion
28  data$TH_branquia_p <- as.numeric(gsub(",", ".", data$TH_branquia_p))
29  data$TH_musculo_p <- as.numeric(gsub(",", ".", data$TH_musculo_p))
30  str(data)
31
32  # Long format
33  th_data <- pivot_longer(
34    data,
35    cols = c(TH_branquia_p, TH_musculo_p),
36    names_to = "Tecido",
37    values_to = "TH"
38  )
39
40  # Calculate the average for each combination of species, Amostra_tipo, and Site
41  average <- summarise(
42    group_by(th_data, Site, Especie, Tecido),
43    Average = mean(TH, na.rm = TRUE),
44    n = sum(!is.na(TH)),

```



```

45   SE = ifelse(
46     n > 1,
47     sd(TH, na.rm = TRUE) / sqrt(n),
48     0
49   ),
50   .groups = "drop"
51 )
52 # Remove groups with no observations
53 average <- average[average$n > 0, ]
54
55 ##### PLOT FINAL #####
56
57 prot_th <-
58   ggplot(
59     average,
60     aes(
61       x = Especie,
62       y = Average,
63       fill = Tecido
64     )
65   ) +
66   geom_col(
67     position = position_dodge(width = 0.9)
68   ) +
69   geom_errorbar(
70     aes(
71       ymin = Average - SE,
72       ymax = Average + SE
73     ),
74     width = 0.2,
75     position = position_dodge(width = 0.9)
76   ) +
77   geom_text(
78     aes(
79       y = 5,
80       label = paste0("", n)
81     ),
82     size = 3,
83     position = position_dodge(width = 0.9)
84   ) +
85   facet_wrap(
86     ~Site,
87     labeller = labeller(

```

```

88     Site = c(
89         "B-P05" = "Site 1",
90         "B-P06" = "Site 2",
91         "B-P07" = "Site 3",
92         "B-P08" = "Site 4",
93         "B-P09" = "Site 5",
94         "B-P10" = "Site 6"
95     )
96 )
97 ) +
98 scale_fill_grey(
99     start = 0.4,
100    end = 0.8,
101    labels = c(
102        "TH_branquia_p" = "Gill",
103        "TH_musculo_p" = "Muscle"
104    )
105 ) +
106 labs(
107     x = "Species",
108     y = "Moisture concentration (%)",
109     fill = "Tissue"
110 ) +
111 theme_grey() +
112 theme(
113     axis.text.x = element_text(
114         angle = 45,
115         hjust = 1
116     ),
117     strip.background = element_rect(
118         fill = "grey85",
119         colour = "grey85"
120     )
121 )
122 prot_th
123 ggsave(prot_th, dpi = 300, filename = "fig-prot_th.png", bg = "white")

1 dev.off()
2 rm(list=ls(all=TRUE))
3 cat("\014")
4
5 data <- read.csv("mxr23_part.csv",
6                 sep = ";", dec = ".",

```

```

7         row.names = 1,
8         header = TRUE,
9         na.strings = NA)
10
11 table(data$Especie, is.na(data$TH_branquia_p))
12 table(data$Especie, is.na(data$TH_musculo_p))
13
14 var <- "TH_musculo_p"
15
16 # Reference site
17 ref <- data[[var]][
18   data$Site == "B-P05" &
19   !is.na(data[[var]])
20 ]
21
22 # Sites to compare
23 sites <- setdiff(unique(data$Site), "B-P05")
24
25 # Run tests
26 results <- data.frame()
27
28 for(site in sites){
29
30   test_data <- data[[var]][
31     data$Site == site &
32     !is.na(data[[var]])
33   ]
34
35   tt <- t.test(
36     test_data,
37     ref,
38     alternative = "greater"
39   )
40
41   results <- rbind(
42     results,
43     data.frame(
44       Site = site,
45       Mean_Site = mean(test_data),
46       SD_Site = sd(test_data),
47       SE_Site = sd(test_data) / sqrt(length(test_data)),
48       Mean_Ref = mean(ref),
49       SD_Ref = sd(ref),

```

```

50     SE_Ref = sd(ref) / sqrt(length(ref)),
51     N_Site = length(test_data),
52     N_Ref = length(ref),
53     t = unname(tt$statistic),
54     df = unname(tt$parameter),
55     p = tt$p.value
56   )
57 )
58 }
59
60 results
61
62 results$Significant <- ifelse(results$p < 0.05, "Yes", "No")
63
64 results$p <- ifelse(
65   results$p < 0.001,
66   "<0.001",
67   sprintf("%.3f", results$p)
68 )
69 #sink("t-tests_prot_th.txt", append = TRUE)
70 results
71 #sink()
72
73 ##### NÃO-PARAMETRICO #####
74
75 results_w <- data.frame()
76
77 for(site in sites){
78
79   test_data <- data[[var]][
80     data$Site == site &
81     !is.na(data[[var])]
82   ]
83
84   wt <- wilcox.test(
85     test_data,
86     ref,
87     alternative = "greater"
88   )
89
90   results_w <- rbind(
91     results_w,
92     data.frame(

```

```

93     Site = site,
94     W = unname(wt$statistic),
95     p = wt$p.value
96   )
97 )
98 }
99
100 results_w$Significant <- ifelse(results_w$p < 0.05, "Yes", "No")
101
102 results_w$p <- ifelse(
103   results_w$p < 0.001,
104   "<0.001",
105   sprintf("%.3f", results_w$p)
106 )
107
108 #sink("t-tests_np.txt", append = TRUE)
109 results_w
110 #sink()

```

MXR phenotype activity. Rhodamine B concentrations differed among sampling sites relative to the reference site (Site 1), which exhibited a mean concentration of $38,396.35 \pm 4,749.86$ RFU mg⁻¹. Significantly higher concentrations were observed at Sites 4, 5 and 6, with mean values of $73,113.69 \pm 5,311.65$, $67,116.05 \pm 7,023.54$ and $62,929.41 \pm 6,196.21$ RFU mg⁻¹, respectively (Wilcoxon rank-sum test; $p < 0.01$). In contrast, Rhodamine B concentrations at Sites 2 and 3 ($16,978.11 \pm 2,139.98$ and $20,973.12 \pm 2,373.55$ RFU mg⁻¹, respectively) were actually significantly lower than those observed at the reference site (two-sided Wilcoxon rank-sum test; $p < 0.01$). The highest mean concentration was recorded at Site 4, corresponding to an approximately 1.9-fold increase relative to the reference site.

```

1 dev.off()
2 rm(list=ls(all=TRUE))
3 cat("\014")
4
5 data <- read.csv("mxr23_part.csv",
6                 sep = ";", dec = ".",
7                 row.names = 1,
8                 header = TRUE,
9                 na.strings = NA)
10
11 table(data$Especie, is.na(data$Rodamina_rfu_mg))
12
13 ##### GRAFICO POR ANÁLISE #####
14

```

```

15 #N #data <- data %>% mutate(N = 1)
16 #RPC
17 #Rodamina_rfu_mg
18 #TH_musculo_p
19 #TH_branquias_p
20 #Proteinas_Totais_mg
21
22 # Convert the column to numeric, replace commas with dots for proper conversion
23 data$Rodamina_rfu_mg <- as.numeric(gsub(",", ".", data$Rodamina_rfu_mg))
24 VAR <- deparse(substitute(data$Rodamina_rfu_mg))
25 var <- "Rodamina_rfu_mg" %>% rlang::sym()
26 str(data)
27
28 # Calculate the average for each combination of species, Amostra_tipo, and Site
29 average <- summarise(
30   group_by(data, Site, Espécie, Amostra_tipo),
31   Average = mean(!var, na.rm = TRUE),
32   n = sum(!is.na(!var)),
33   SE = ifelse(
34     sum(!is.na(!var)) > 1,
35     sd(!var, na.rm = TRUE) / sqrt(sum(!is.na(!var))),
36     0
37   ),
38   .groups = "drop"
39 )
40 average <- average[
41   average$Amostra_tipo == "rd" & !is.nan(average$Average),
42 ]
43 # Plot the average
44 ggplot(average, aes(x = Espécie, y = Average, fill = Amostra_tipo)) +
45   geom_bar(stat = "identity", position = position_dodge()) +
46   facet_wrap(~ Site) +
47   labs(title = paste("Average POR ANÁLISE para", VAR, "per Species and Amostra_tipo for Each",
48     x = "Species",
49     y = paste("Average", VAR)) +
50   theme_minimal() +
51   theme(axis.text.x = element_text(angle = 45, hjust = 1))
52
53 ##### PLOT FINAL #####
54
55 rodam <-
56   ggplot(average, aes(x = Espécie, y = Average, fill = Amostra_tipo)) +
57   geom_bar(stat = "identity", position = position_dodge(), fill = "grey70") +

```

```

58     geom_errorbar(
59     aes(
60         ymin = Average - SE,
61         ymax = Average + SE
62     ),
63     width = 0.2
64 ) +
65 geom_text(
66     aes(
67         y = 5000,
68         label = paste0("n=", n)
69     ),
70     size = 3
71 ) +
72 facet_wrap(
73     ~ Site,
74     labeller = labeller(
75         Site = c(
76             "B-P05" = "Site 1",
77             "B-P06" = "Site 2",
78             "B-P07" = "Site 3",
79             "B-P08" = "Site 4",
80             "B-P09" = "Site 5",
81             "B-P10" = "Site 6"
82         )
83     )
84 ) +
85 scale_fill_grey(start = 0.2, end = 0.8) +
86 labs(title = "",
87     x = "Species",
88     y = "Rhodamine B concentration (RFU/mg tissue)",
89     fill = "Species") +
90 theme_grey() +
91 theme(axis.text.x = element_text(angle = 45, hjust = 1),
92     legend.position = "none")
93 rodam
94 ggsave(rodam, dpi = 300, filename = "fig-rodam.png", bg = "white")

1 dev.off()
2 rm(list=ls(all=TRUE))
3 cat("\014")
4
5 data <- read.csv("mxr23_part.csv",

```

```

6         sep = ";", dec = ".",
7         row.names = 1,
8         header = TRUE,
9         na.strings = NA)
10
11 table(data$Especie, is.na(data$Rodamina_rfu_mg))
12 var <- "Rodamina_rfu_mg"
13
14 # Reference site
15 ref <- data[[var]][
16   data$Site == "B-P05" &
17   !is.na(data[[var])]
18 ]
19
20 # Sites to compare
21 sites <- setdiff(unique(data$Site), "B-P05")
22
23 # Run tests
24 results <- data.frame()
25
26 for(site in sites){
27
28   test_data <- data[[var]][
29     data$Site == site &
30     !is.na(data[[var])]
31 ]
32
33   tt <- t.test(
34     test_data,
35     ref,
36     alternative = "greater"
37   )
38
39   results <- rbind(
40     results,
41     data.frame(
42       Site = site,
43       Mean_Site = mean(test_data),
44       SD_Site = sd(test_data),
45       SE_Site = sd(test_data) / sqrt(length(test_data)),
46       Mean_Ref = mean(ref),
47       SD_Ref = sd(ref),
48       SE_Ref = sd(ref) / sqrt(length(ref)),

```



```

49     N_Site = length(test_data),
50     N_Ref = length(ref),
51     t = unname(tt$statistic),
52     df = unname(tt$parameter),
53     p = tt$p.value
54 )
55 )
56 }
57
58 results
59
60 results$Significant <- ifelse(results$p < 0.05, "Yes", "No")
61
62 results$p <- ifelse(
63     results$p < 0.001,
64     "<0.001",
65     sprintf("%.3f", results$p)
66 )
67 #sink("t-tests_rodam.txt", append = TRUE)
68 results
69 #sink()
70
71 ##### NÃO-PARAMETRICO #####
72
73 results_w <- data.frame()
74
75 for(site in sites){
76
77     test_data <- data[[var]][
78         data$Site == site &
79         !is.na(data[[var])]
80     ]
81
82     wt <- wilcox.test(
83         test_data,
84         ref,
85         alternative = "two.sided"
86     )
87
88     results_w <- rbind(
89         results_w,
90         data.frame(
91             Site = site,

```

```

92     W = unname(wt$statistic),
93     p = wt$p.value
94   )
95 )
96 }
97
98 results_w$Significant <- ifelse(results_w$p < 0.05, "Yes", "No")
99
100 results_w$p <- ifelse(
101   results_w$p < 0.001,
102   "<0.001",
103   sprintf("%.3f", results_w$p)
104 )
105
106 #sink("t-tests_np.txt", append = TRUE)
107 results_w
108 #sink()

```

4 Discussion

The present study evaluated whether biomarkers of physiological stress in headwater stream fish could detect environmental disturbances occurring within and around a protected area of Atlantic Forest in northeastern Brazil. Despite sites within the administrative boundaries of the REBIO showing no signs alteration, from both environmental variables and biomarkers, significant differences in biomarker responses were observed among sampling sites, indicating that environmental conditions are not homogeneous throughout the stream studied. Elevated levels of all biomarkers at sites located outside or near the boundaries of the protected area suggest exposure to environmental stressors potentially associated with anthropogenic activities. These findings support the use of biochemical and physiological biomarkers as sensitive tools for detecting sublethal effects of environmental degradation, even in landscapes that contain legally protected areas.

stress levels occurring inside and outside of a Conservation Unity was tested in the Guaribas Environmental Reserve (REBIO Guaribas), one of the 22 CU's inside the Atlantic forest area of Paraíba state, located in the Zona da Mata, wettest and most economically relevant mesoregion from the state (Figure 1). Grouped, according to SNUC, as one of the integral protection areas (aimed to conserve biodiversity, being minority within the biogeographical province – 33,45% of the CU's), the definition of environmental reserves prohibits human interference inside their limits, except for management actions that aim its preservation and recovery. Despite being one of the most restricted types of Conservation Units, there are 22 populations that influence REBIO Guaribas due to historic factors of occupation and absence of an effective buffer zone around the three areas that make up the reserve. Those populations are mainly

rural and cause direct impact, by using wood and plants from the reserve, and indirect, causing siltation of rivers and contaminating groundwater through the use of septic tanks and releasing agricultural chemicals into the soil, promoting possible comparable ecophysiological responses in the ichthyofauna inside and around the CU (Arruda et al., 2013; MMA, 2026).

The ecophysiological evaluation of human impact on the forest, which is close to areas of monoculture and natural vegetation modified for livestock, was mainly guided by the fish species inventory of REBIO Guaribas made by Gouveia et al. (2017), in a study that lists 18 species from five different orders: Characiformes (12 species), Perciformes (3 species), Synbranchiformes (1 species), Siluriformes (1 species) e Cyprinodontiformes (1 species). From those, two are exotic - *Oreochromis niloticus*, second most farmed fish species in the world and tolerant to osmotic variation, and *Poecilia reticulata*, largely used in mosquito control and resistant to stressful environments (Miranda, 2012) - and none are listed as threatened. The predominance of small fish inhabiting small basins explains the higher level of endemism in streams, resulting from the allopatric speciation generated by the low dispersion made by these fish and the consequent isolation of populations (Gouveia et al., 2017). Having in mind that these stream fish are phylogenetic close, this paper offers representative data for the entire local ichthyofauna using species of *Hemigrammus* genus, of the Characiformes order, characterized by a black conspicuous spot delimited on the dorsal fin. Its type species, *Hemigrammus unilineatus*, has a compressed and slightly elongated body and exhibits wide geographical distribution both in Central and South America, with registers in other Brazilian states, such as Alagoas and Pernambuco.

Biomarkers. The increase of total plasmatic protein concentration in the animals exposed to pollution is consistent with the (Sabae & Mohamed, 2015) study, which attributes this increase to five possibilities: activation of metabolic systems as answer to pollutants exposure, degradation of cellular material in the liver, severe pathological conditions in the liver and kidneys, loss of water in the plasma, and induction of proteic synthesis in the liver. The average values of moisture content, parameter related to the osmoregulatory function, were 84,3% for the gill and 80,3% for the muscle of fish inside the REBIO Guaribas (n=38 fish). Regarding the fish from sampling sites surrounding the reserve (n=40), the averages were 88,9% for the gills and 82,3% for the muscle, increasing, in order, only about 5,6% and 2,5% the values from inside, results that are not very expressive, like the ones described by Haredi et al. (2020). The average values of moisture content, parameter related to the osmoregulatory function, were 84,3% for the gill and 80,3% for the muscle of fish inside the REBIO Guaribas (n=38 fish). Regarding the fish from sampling sites surrounding the reserve (n=40), the averages were 88,9% for the gills and 82,3% for the muscle, increasing, in order, only about 5,6% and 2,5% the values from inside, results that are not very expressive, like the ones described by Haredi et al. (2020). The overall statistical analysis performed for moisture content reinforces this as a more discreet and inefficient parameter to make direct quantitative comparison for animals exposed to low salinity variation from the environment, but still accurate in determining the uniformity degree of more representative groups (i.e., fish from inside and outside the Conservative Unity). The average values of normalized fluorescence measured in fish gills were 25.449 rfu/mg (relative fluorescence units/milligramme of tissue) for inside the Conservative

Unity (n=54 fish) and 67.138,3 rfu/mg for outside (n=57), value nearly 2,64 times bigger. This difference between averages is opposite to the expected (higher fluorescence signals a lower MXR phenotype activity, meaning a lower exposition to pollutants), that can be explained by the presence of MXR phenotype inhibitors, which were highly correlated to more polluted waters by Kurelec et al. (2000). Among the possible pollutants presents, pesticides, fragrances, microbial degradation products and natural inhibitors from invasive species were described as compounds with high affinity to the Pgp or inhibiting the PKC (protein kinase C) modulator, resulting in high accumulation of rhodamine B (Smital et al., 2004). Despite the unexpected result, the variance difference between data from the MXR phenotype activity inside and outside the REBIO Guaribas (confirmed by the Kruskal-Wallis test and illustrated in *The abiotic data from every sampling site water was measured and reunited in Tab. 1. The abnormal proportion of ammonia indicates the presence of effluents in the water outside the Reserva Biológica Guaribas, and, as discussed by (Piedras et al., 2006), this elevated concentration generates metabolic ammonia retention, causing toxicity. The same paper also points out that the toxicity of non-ionized ammonia in the aquatic environment is greater for smaller organisms, but depends on the interaction with other environmental variables, such as temperature, pH and salinity, demanding specific studies with the target species to determine their tolerance to ammonia. The Pearson correlation analysis among biotic data showed only one significant correlation, between Total Protein and MXR phenotype activity (0,6 - moderate and positive), and among abiotic data only two significance, between Nitrate e Phosphor (0,69 - moderate and positive) and between Dissolved Oxygen and Ammonia (-0,82 - strong and negative). It was also detected significant correlation among both data, between Gill Moisture Content and Ammonia (0,55 - moderate and positive), between Gill Moisture Content and Dissolved Oxygen (-0,62 - moderate and negative), and between MXR activity and Nitrate (-0,62 - moderate and negative).* The elevated ammonia concentrations observed at Sites 4–6, particularly at Site 6, may indicate inputs of domestic or agricultural effluents into the aquatic environment. According to Piedras et al. (2006), increased environmental ammonia can lead to metabolic ammonia retention and toxic effects in fish, although toxicity depends on interactions with other environmental variables such as temperature, pH, and salinity. The strong negative correlation between dissolved oxygen and ammonia concentration observed in the present study further suggests that sites with greater organic inputs may be experiencing water quality degradation.

All parameters were effective to point out statistical differences between data from inside the REBIO Guaribas and its surroundings, highlighting - alongside the higher levels of ammonia in the water and plasmatic proteins in the fish - that changes in the environment are impacting the fish from the streams studied. Beyond that, the statistical analysis of physiological parameters from fish collected inside the Conservation Unity showed uniform results among its three sampling sites, consistent with what is expected from natural environments with little human influence. The same did not occur among the three sites outside the CU (with the exception of the MXR phenotype activity), indicating the action of different levels of environmental stress. The differences between the average results from the physiological parameters measured inside and outside the Conservation Unity were illustrated in the fig-bar, highlighting the total

protein as one that proved itself as an excellent biomarker to detect pollutants in the aquatic environment.

These preliminary obtained data were precise to indicate significant differences among the conditions in which the fish inhabited. But as much MXR phenotype activity data were as disparate as the ones of total protein and even more precise to point out the uniformity among sampling sites inside and outside the REBIO Guaribas, it showed values opposite to what was expected, with higher fluorescence in the tissue of fish outside the reserve, normally indicating lower MXR phenotype activity. This inconsistency can be explained by the possible presence of chemosensitizers in the water, since the fish did not went through a decontamination period, and thus, pollutants could still be bound to the active Pgp site and preventing rhodamine B to attach to it. The presence of such chemosensitizers can be quantified through the creation of a calibration curve, as described by (Kurelec et al., 2000), but, until done, the relevance of the MXR phenotype analysis must be restricted to the statistical context.

From the final results, an important positive evaluation can be made regarding the environmental protection that the Conservative Unity does within its extension, even when analysing one with historical problems of delimitation and still not entirely free of anthropic impacts. Adding the fact that it is located within the Atlantic forest, classified as a Hotspot, the verification of the efficiency of a CU in this biogeographical province is even more relevant, reinforcing the need for governments to create more legally protected areas as a key strategy to reduce global ecological crisis. This study also made an important description on the use of physiological biomarkers in ecological analysis, raising ecophysiology potential as a basis to future assays, which can be incorporated into effectiveness index such as seen in (Masullo et al., 2020) paper and presenting quantifiable data that can complement studies with analysis of historical and geographical data, such as the one from (Assis et al., 2021).

Conclusions

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Authorship contribution statement

Authorship of this paper is based on CRediT (2026).

Adamastor Coutinho Pinto: Investigation, Formal analysis, Writing – original draft.

Larissa Rafaela Caetano da Silva: Data curation, Formal analysis.

Mayara Mirelly da Silva Monteiro: Data curation, Formal analysis, Software, Writing – review & editing.

Alice da Silva Barros: Data curation, Formal analysis, Software, Writing – review & editing.

Enelise Marcelle Amado: Data curation, Methodology, Resources, Supervision, Validation, Writing – review & editing.

Elvio Sergio Figueredo Medeiros: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Validation, Writing – review & editing.

Ethical approval

Fish collection was authorized by the Instituto Chico Mendes de Conservação da Biodiversidade (license SISBIO-89415-2) and the project was approved by the Ethics Committee on Animal Use of the State University of Paraíba (protocol CEUA-048/2023).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to assist with English language editing and improve the clarity and fluency of the writing. The authors carefully reviewed and revised all AI-generated suggestions and take full responsibility for the accuracy and content of the final manuscript (CNPq, 2026).

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Figures and Tables

Figure 1: Location of sampling sites along the Barro Branco Stream (Mamanguape, PB) (Site 1 06°43'06"S and 35°10'54"W; Site 2 06°42'38"S and 35°10'38"W; Site 3 06°43'06"S and 35°10'54"W; Site 4 06°41'46"S and 35°10'36"W; Site 5 06°40'56"S and 35°10'27"; Site 6 06°40'18"S and 35°10'34"W). Site 1 is the reference site for this study.

Figure 2: Images of the main four types of sites found along the Barro Branco stream (Mamanguape, PB). Type I, well preserved and within the conservation unit. Type II site, preserved and outside the conservation unit. Type III site, disturbed but withing the conservation unit. Type IV site, disturbed and outside the conservation unit.

Figure 3: Principal Component Analysis of the environmental variables for the Barro Branco stream (Mamanguape, PB).

Figure 4: Counts of number of individuals per species and biomarker use (MXR, Multixenobiotic resistance phenotype activity; Moisture, Plasmatic protein dosage and moisture content) for each study site.

Figure 5: Total plasmatic proteins (mg) variance by sampling site and fish species.

Figure 6: Gill and muscle moisture content (%) by sampling site and fish species.

Figure 7: Average fluorescence of rhodamin B expressed by relative fluorescence units/milligrams of gill tissue of fish by sampling site.

Table 1: Environmental variables averaged for sampling sites (minimum-maximum) for the Barro Branco stream (Mamanguape, PB). Site 1 is the reference site for this study.

Table 2: Nutrient concentration (g L⁻¹) averaged for sampling sites (minimum-maximum) for the Barro Branco stream (Mamanguape, PB). Site 1 is the reference site for this study.

Figures

Fig.: Location of sampling sites

Fig.: Images of the main four types of sites

Fig.: Principal Component Analysis of the environmental variables

Fig.: Counts of number of individuals per species and biomarker

Fig.: Total protein variance

Fig.: Moisture content of tissues

Fig.: Average fluorescence for rhodamin B

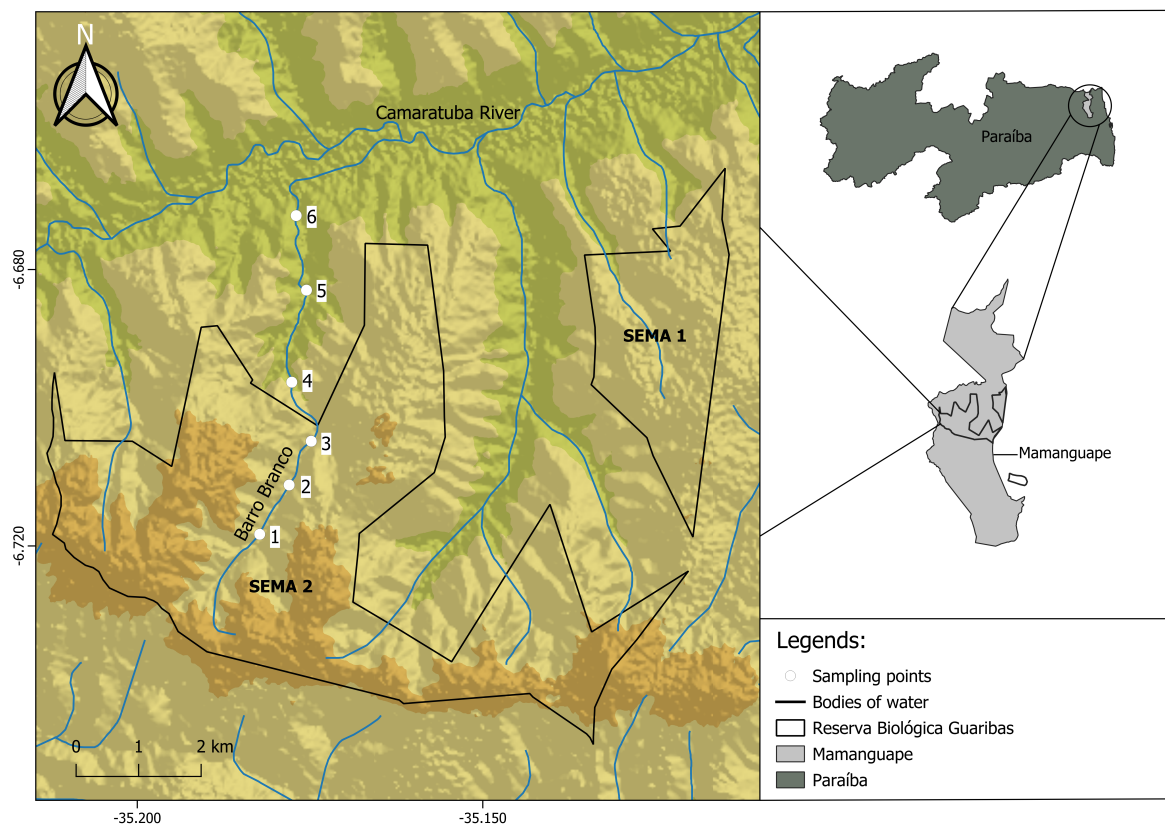


Figure 1: Location of sampling sites.

Type I



Type II



Type III



Type IV



Figure 2: Images of the main four types of sites.

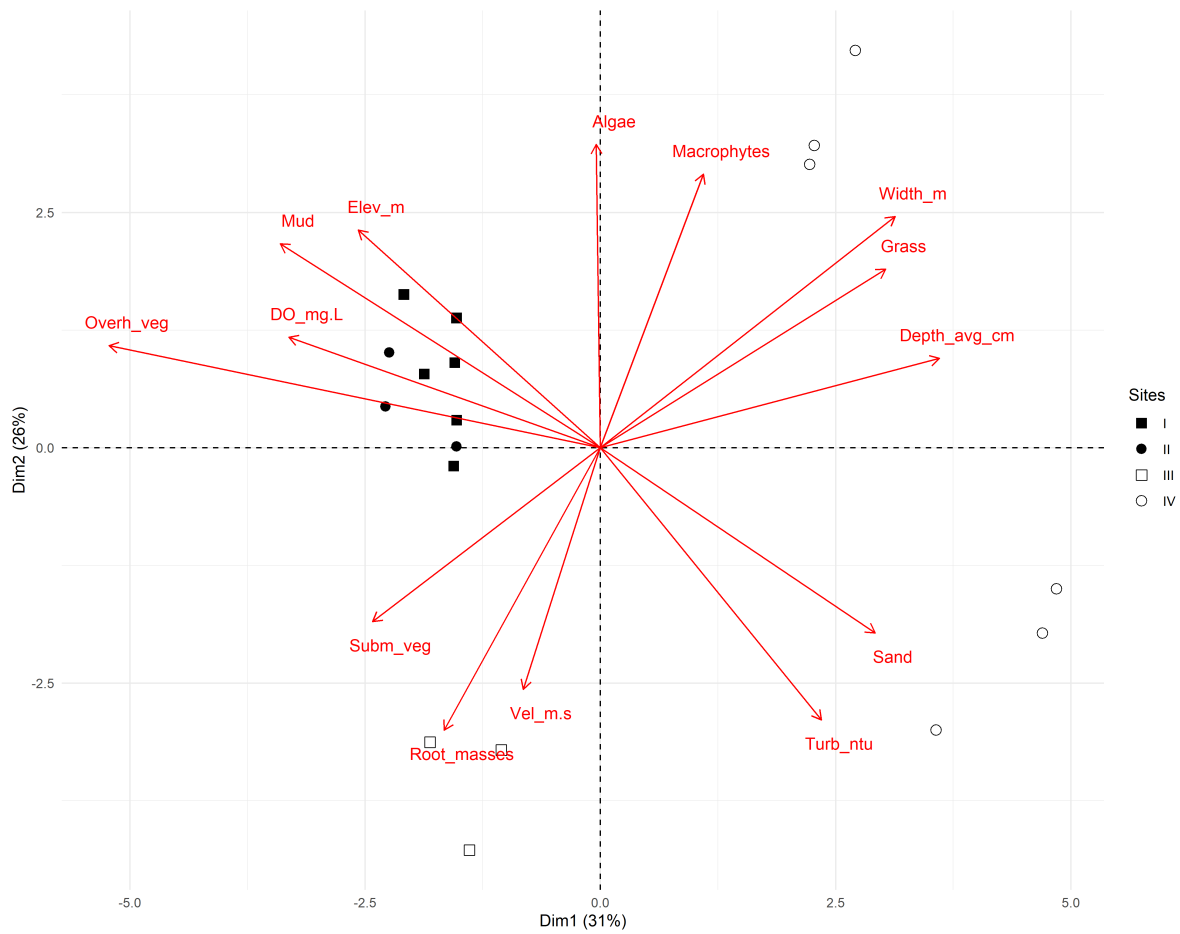


Figure 3: Principal Component Analysis of the environmental variables.

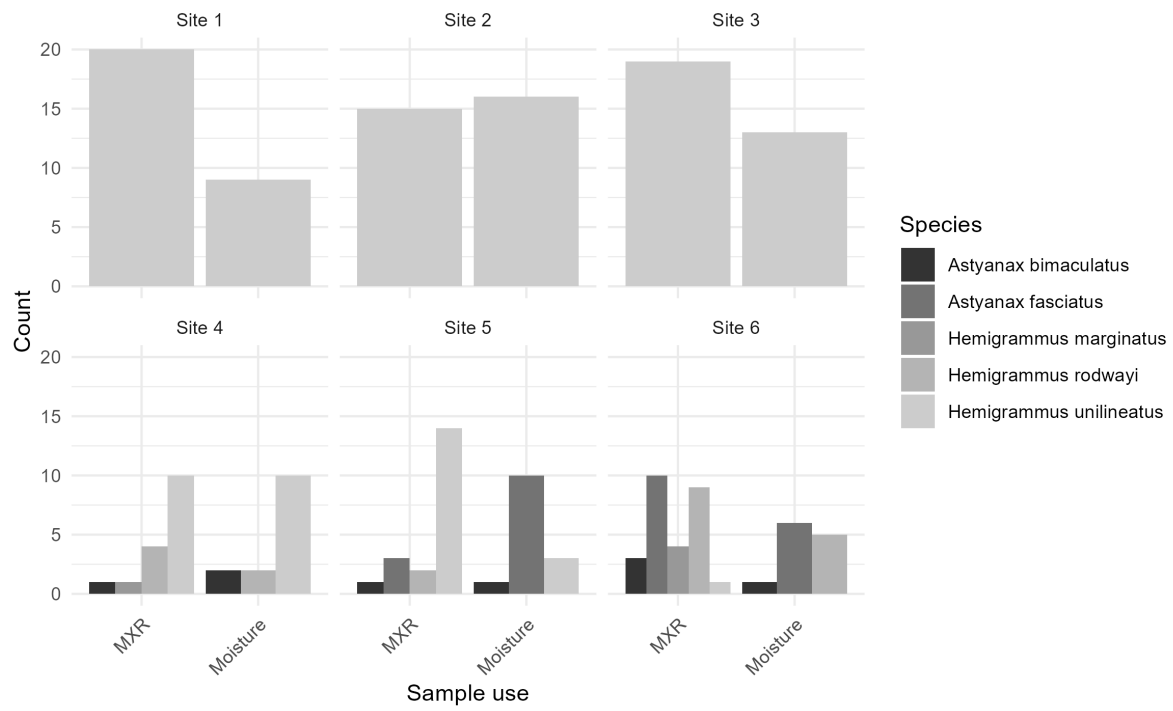


Figure 4: Counts of number of individuals per species and biomarker.

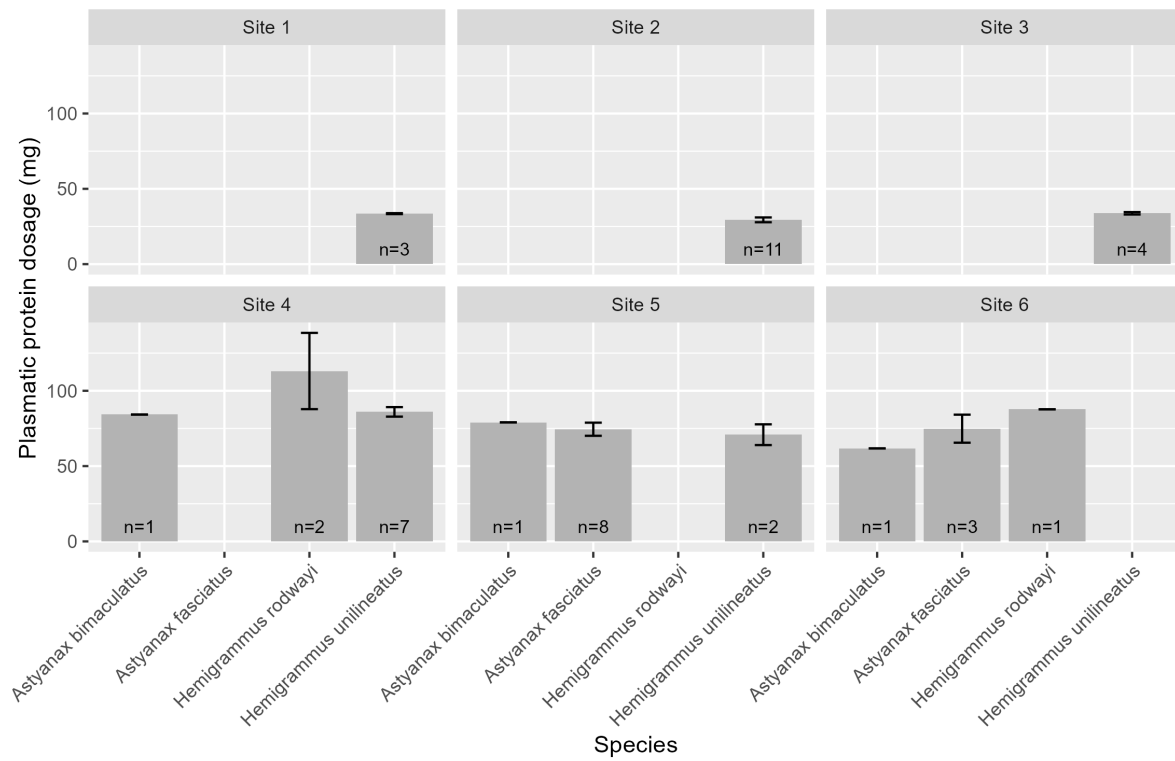


Figure 5: Total plasmatic proteins variance.

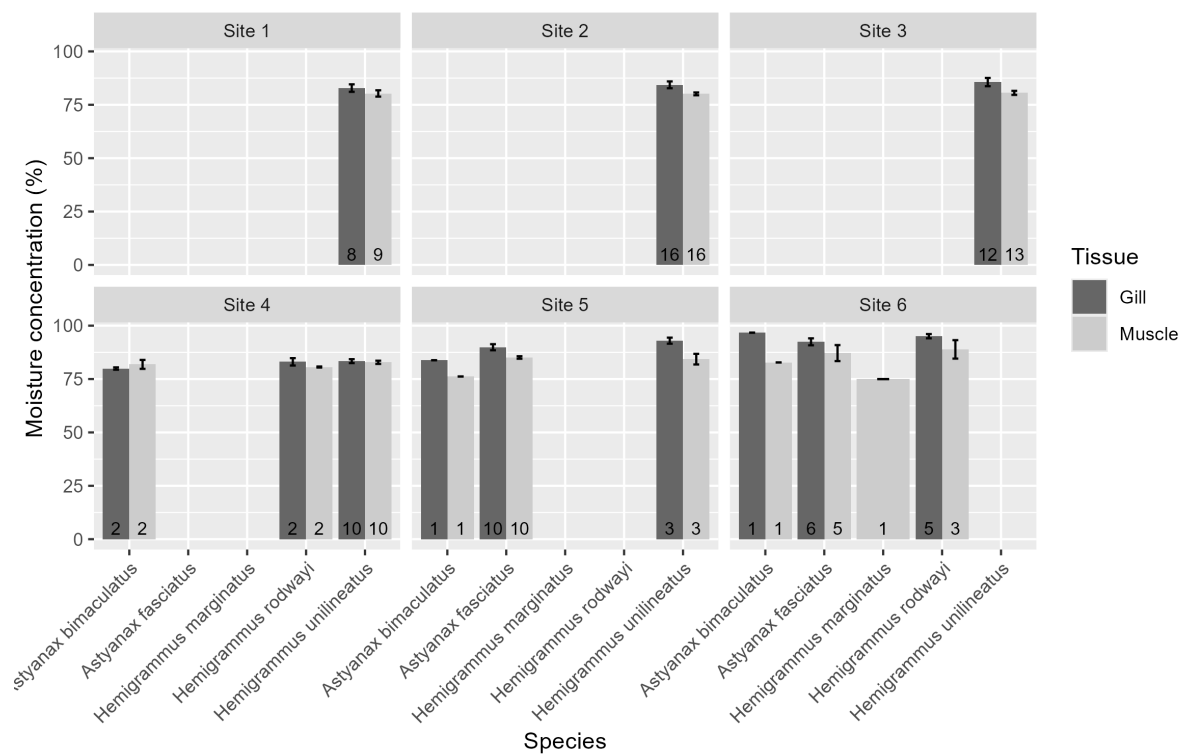


Figure 6: Gill and muscle moisture content.

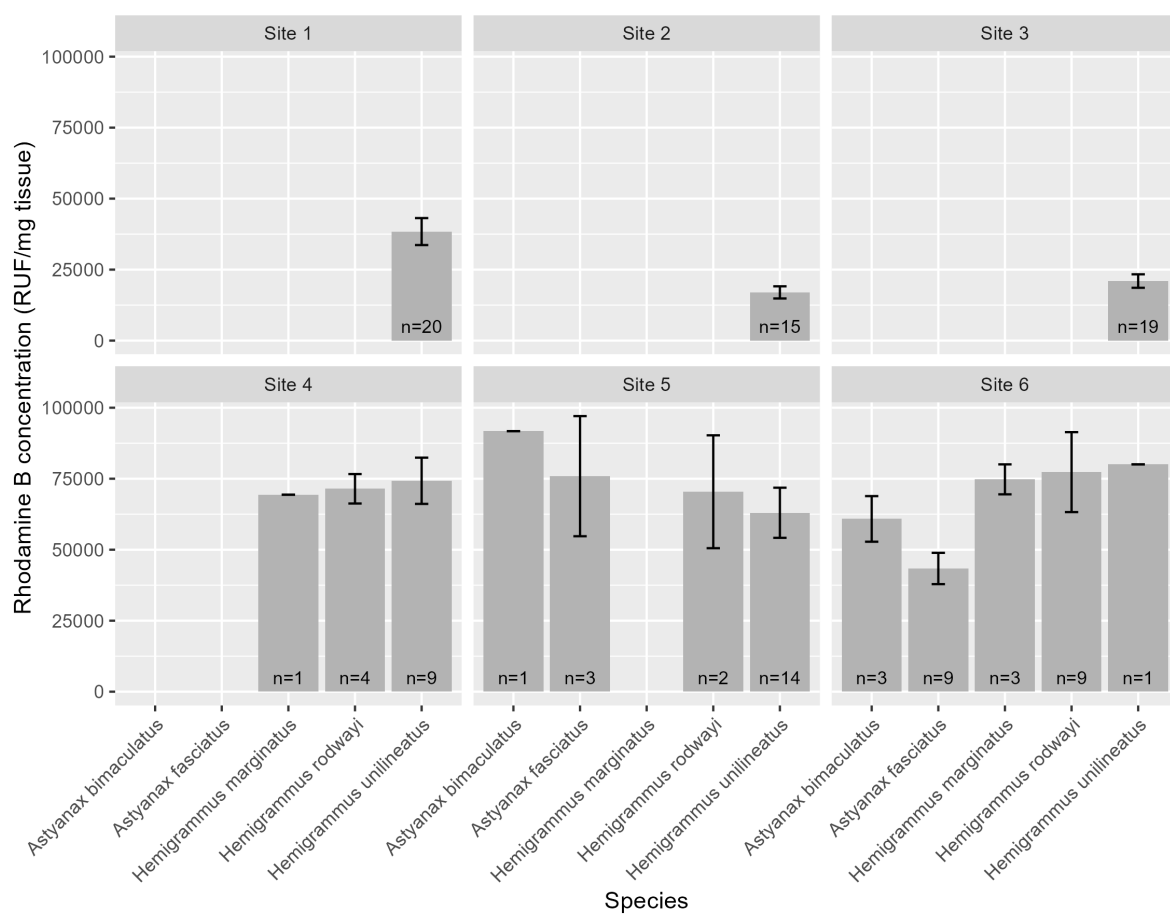


Figure 7: Average fluorescence of rhodamin B.

Table 1: Environment data GT table.

Variable	Site 1	Site 2	Site 3	
Water quality				
Temperature (°C)	25.8 (24.9-27.3)	26.5 (25.3-27.5)	25.2 (24.9-25.5)	27.8
Dissolved oxygen (mg/L)	8.1 (7.8-8.5)	7.5 (7.2-7.8)	7.9 (7.7-8)	7.5
Conductivity (uS/cm)	146.4 (134-164.4)	148.3 (134.1-156.8)	129 (98.3-160.2)	160.6 (158)
Turbidity (NTU)	7.4 (4.6-10.7)	6.1 (2.1-14.1)	4.4 (1.5-9.2)	6.4
pH	7 (6.3-7.5)	6.8 (5.9-7.7)	5.5 (4.6-6)	5.
Morphology				
Elevation (m a.s.l.)	107.1 (107.1-107.1)	112.8 (112.8-112.8)	89 (89-89)	89.1 (8)
Width (m)	3.6 (3.6-3.6)	3.7 (2.3-4.4)	1.7 (1.1-2.1)	14.6 (1)
Water velocity (m/s)	0.017 (0-0.025)	0.035 (0.035-0.035)	0.073 (0.022-0.108)	
Slope	1.2 (1-1.5)	2.5 (2.5-2.5)	2.2 (1.5-2.5)	
Average depth (cm)	13.7 (13.7-13.7)	23 (20.3-28.3)	19.7 (16-22.7)	44.2
Substrate composition (%)				
Mud	100 (100-100)	100 (100-100)	90 (90-90)	80
Sand	0 (0-0)	0 (0-0)	10 (10-10)	20
Habitat structures (%)				
Aquatic macrophytes	5 (5-5)	20 (0-30)	0 (0-0)	35.8
Littoral grass	0 (0-0)	5 (5-5)	2 (0-5)	39.2
Submerged vegetation	5 (5-5)	8.3 (5-15)	3.3 (0-10)	
Overhanging vegetation	90 (90-90)	81.7 (80-85)	85 (80-90)	30
Leaf litter	10 (10-10)	3.7 (1-5)	20 (5-30)	26.7
Algae	5 (5-5)	0.3 (0-1)	11 (2-21)	16.7
Debris	30 (30-30)	5.3 (5-6)	10.3 (5-15)	31.7
Root masses	0 (0-0)	8.3 (5-15)	10 (5-15)	

Tables

Tab.: Environmental data table

Tab.: Nutrient data table

Table 2: Nutrient concentrations at the sampling sites.

Variable	Site 1	Site 2	Site 3
Ammonia ($\mu\text{g L}^{-1}$)	54.773 (39.32-64.32)	58.867 (47.5-77.05)	72.047 (33.41-115.68)
Nitrite ($\mu\text{g L}^{-1}$)	0 (0-0)	0 (0-0)	0 (0-0)
Nitrate ($\mu\text{g L}^{-1}$)	6 (0-9.7)	8.8 (6.9-12.6)	5.5 (0-12.6)
Orthophosphate ($\mu\text{g L}^{-1}$)	0 (0-0)	0 (0-0)	1 (0-3)
Phosphorus ($\mu\text{g L}^{-1}$)	0 (0-0)	4.6 (0.3-10.3)	50.1 (0-150.3)

Apêndices

Non-used figures and tables

⚠ Non-used codes

```

1  table(1:10)
1  library(readr)
2  library(dplyr)
3  library(gt)
4
5  m_wide_hab <- read.table("m_wide_hab.txt",
6                          sep = ";", dec = ".",
7                          row.names = 1,
8                          header = TRUE,
9                          na.strings = NA)
10
11 #m_mapa <- read_tsv("column_labels.txt", show_col_types = FALSE)
12 dados <- m_wide_hab
13 dados
14 nomes <- names(dados)[-1]
15 nomes
16
17 df_nomes <- data.frame(
18   original = nomes,
19   final     = nomes,
20   stringsAsFactors = FALSE)
21
22 df_nomes
23 df_nomes$final <- gsub("_", "<br>", df_nomes$final)

```

```

24 dados <- mutate(dados, across(-Variable, ~ gsub(" \\(", "<br>(", .)))
25 #fix(df_nomes)
26 #write.table(df_nomes, "df_nomes.txt")
27 df_nomes <- read.table("df_nomes.txt")
28
29 labels_finais <- setNames(
30   lapply(df_nomes$final, md),
31   nomes)
32
33 tabela_gt <- dados %>%
34   gt(rowname_col = "Variable") %>%
35   # tab_header(
36   #   title = "Características Ambientais",
37   #   subtitle = "Valores apresentados como média (mín-máx)"
38   # ) %>%
39   fmt_markdown(
40     columns = -Variable) %>%
41   cols_align(
42     align = "right",
43     columns = -Variable) %>%
44   cols_label(.list = labels_finais)
45
46 tabela_gt
47 gtsave(tabela_gt, "gt-hab_gtable.html")
48 gtsave(tabela_gt, "fig-hab_gtable.png")
49 saveRDS(tabela_gt, "gt-hab_gtable.rds")

```